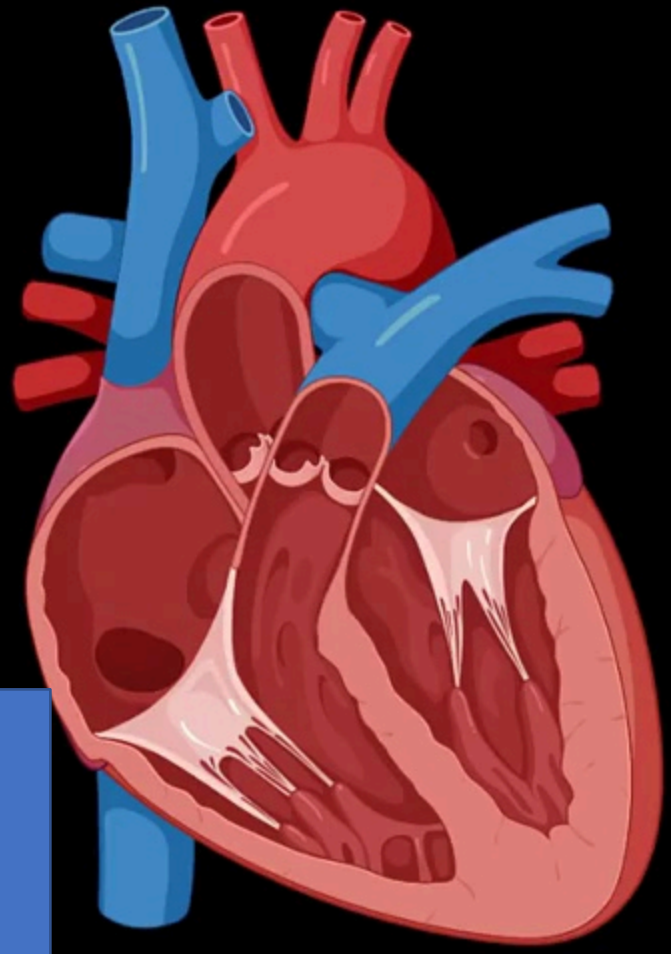
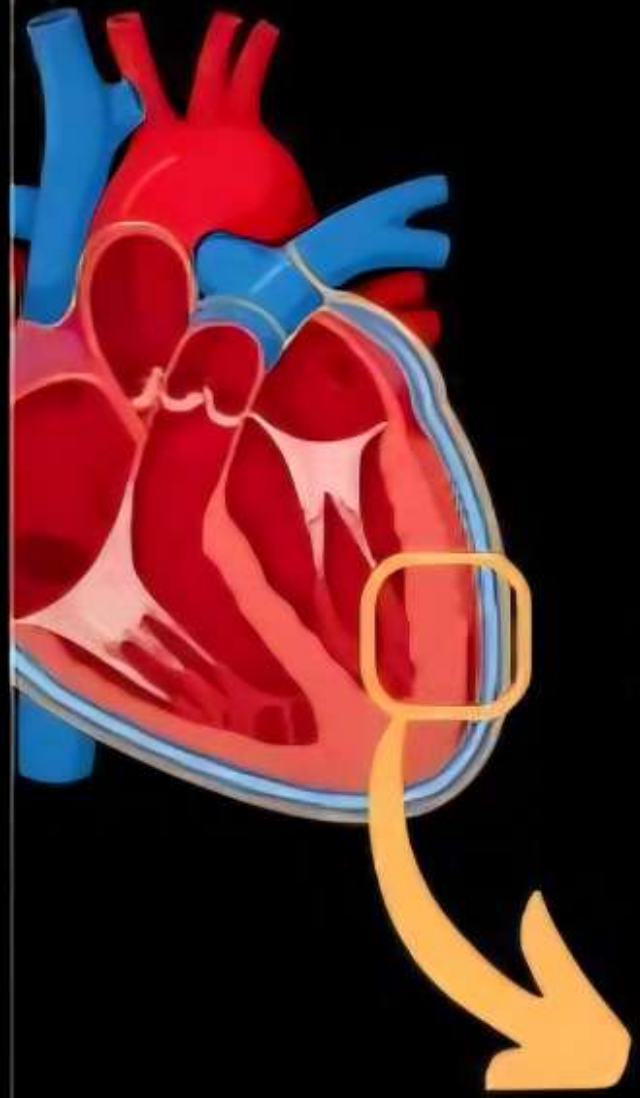
Heart Anatomy

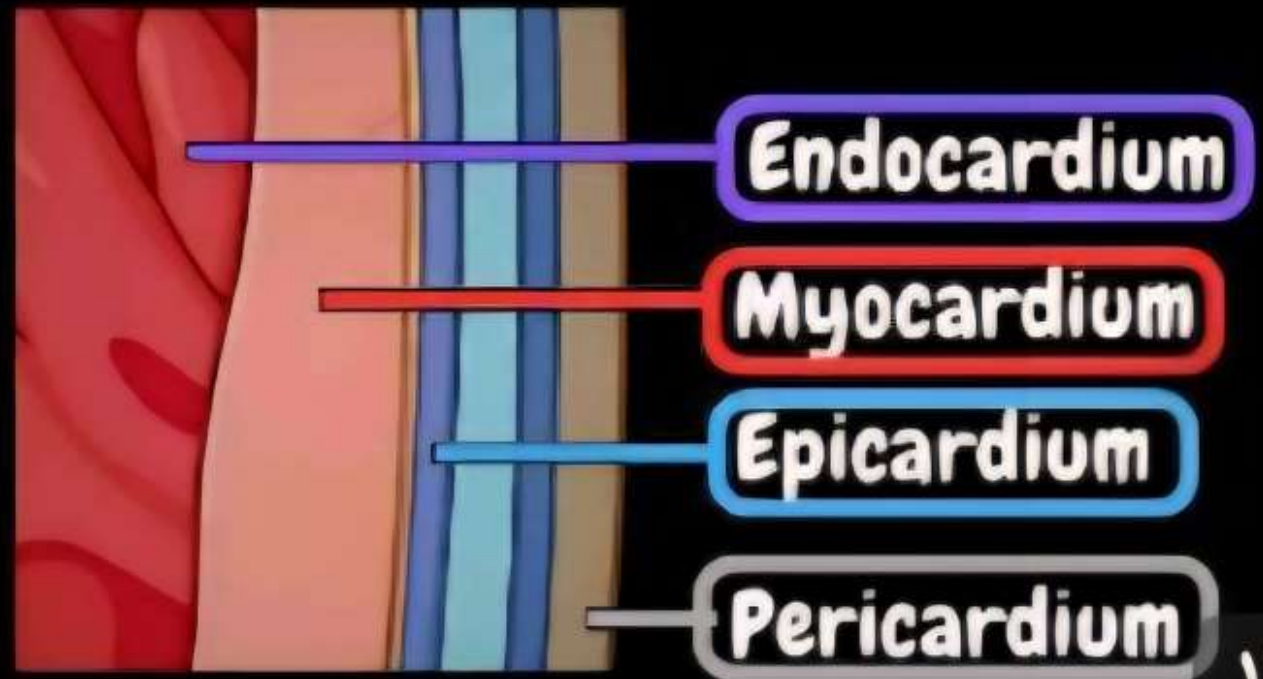


Dr.m.almusaibli

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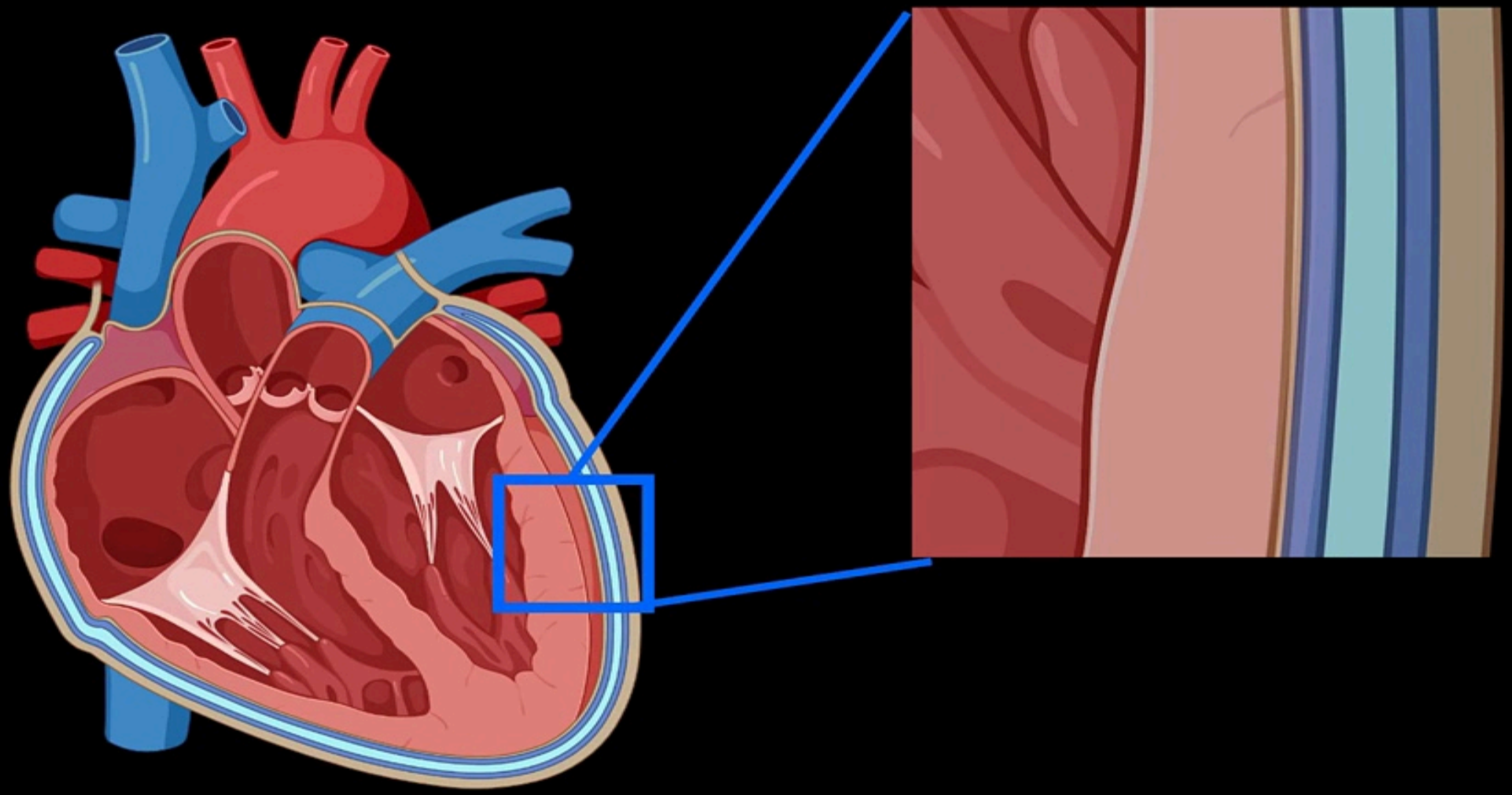
Heart Layers



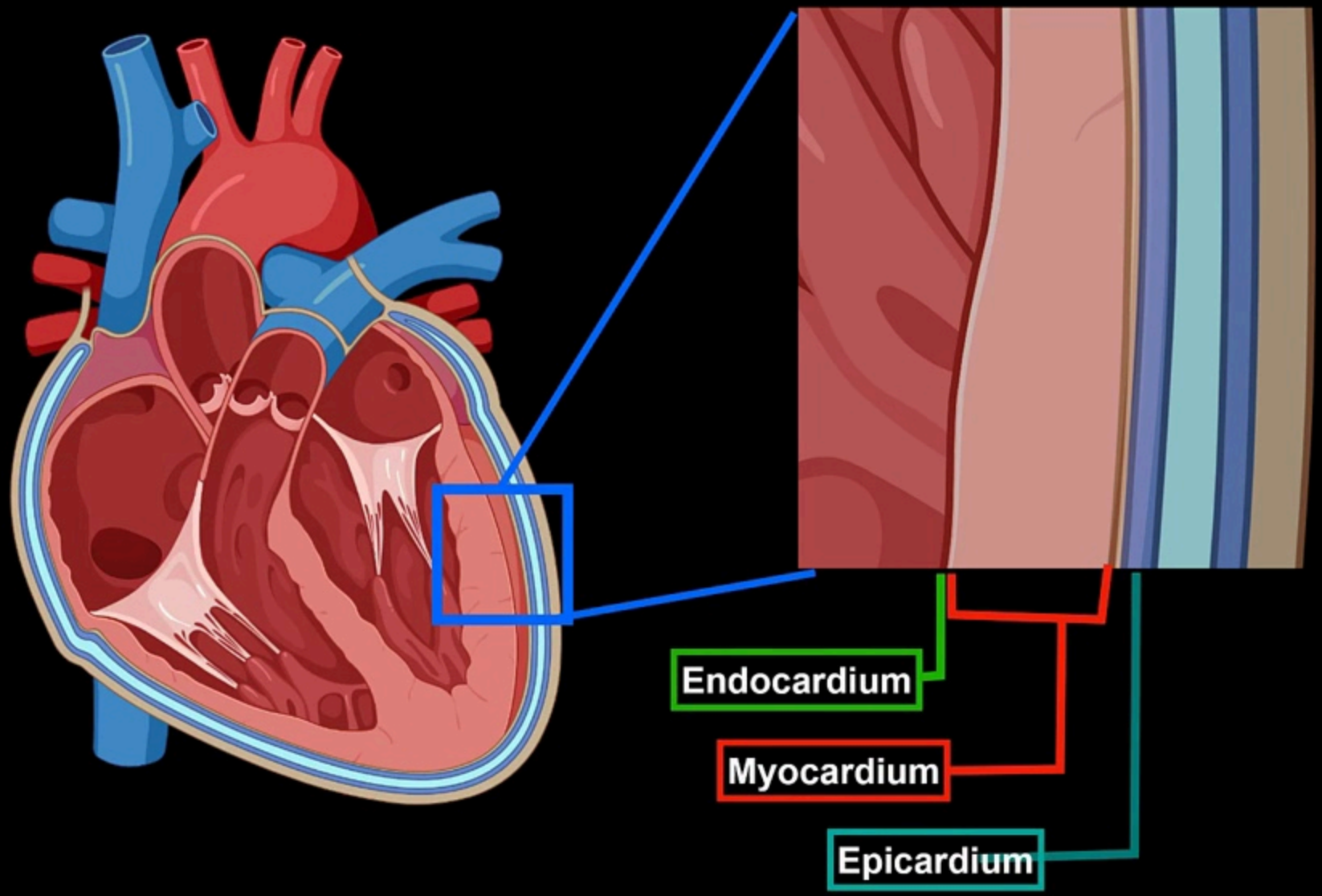


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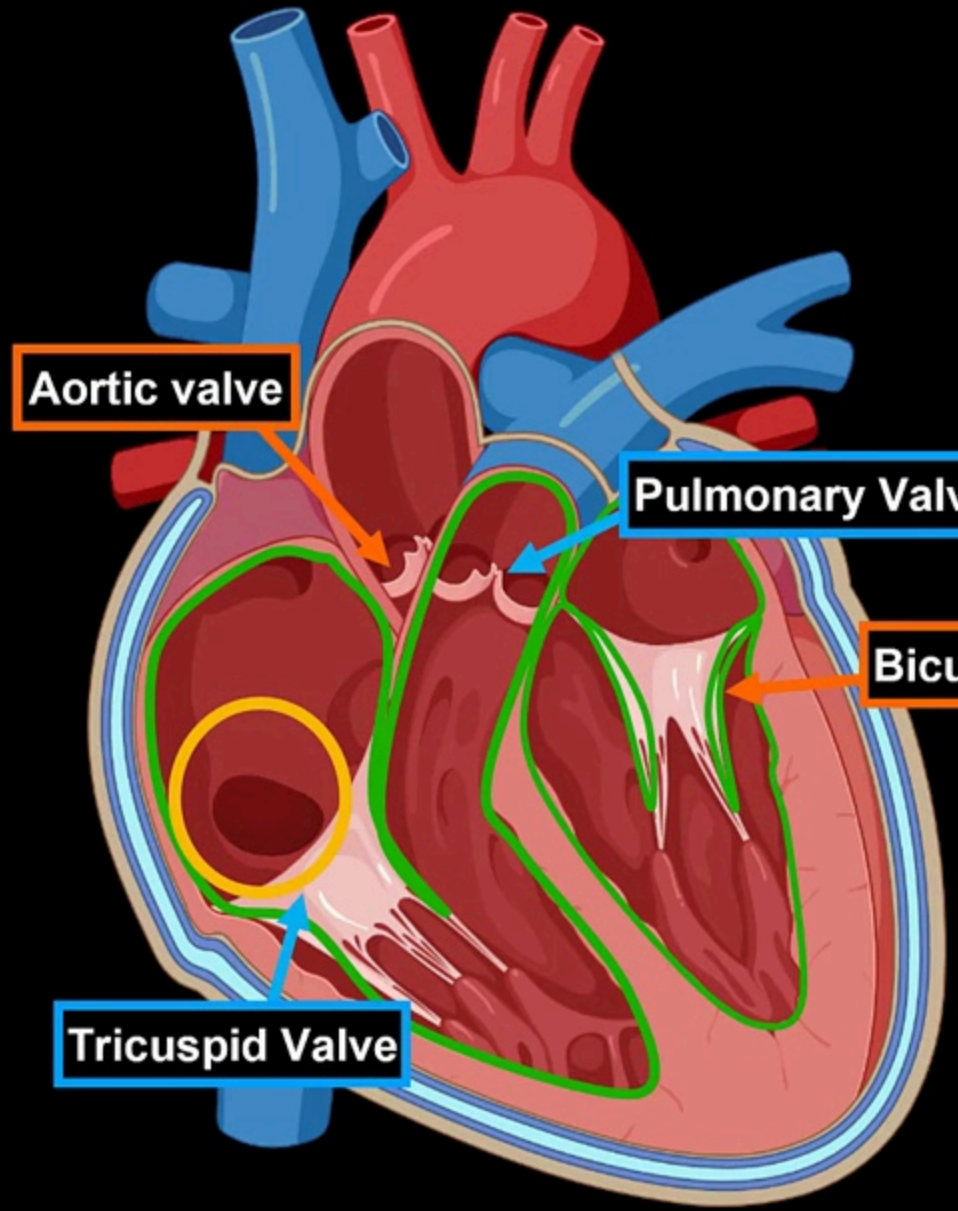
Layers of the Heart



Layers of the Heart



Endocardium

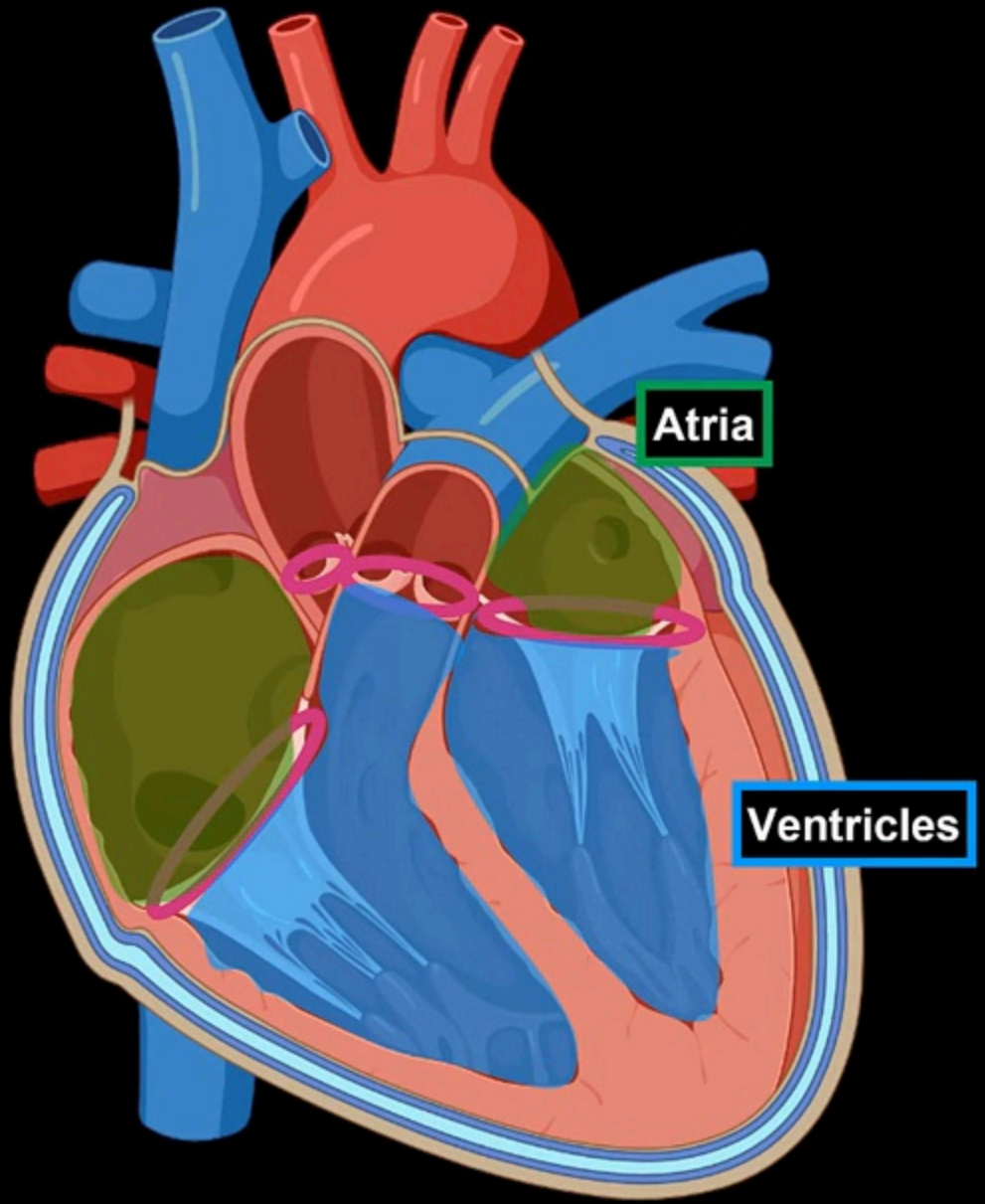


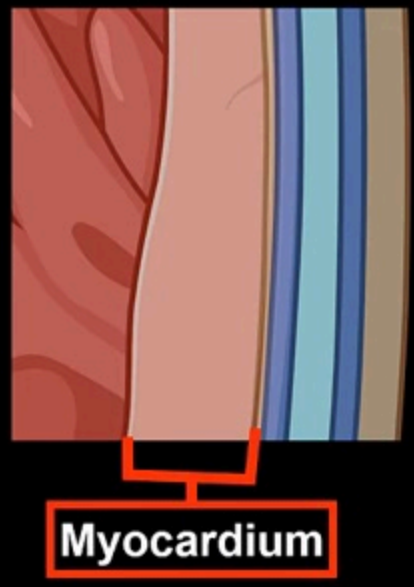


Valve of the Inferior Vena Cava
Vulva Vena Cava Inferioris

Valve for the Coronary Sinus
Vulva Sinus Coronarii

Myocardium

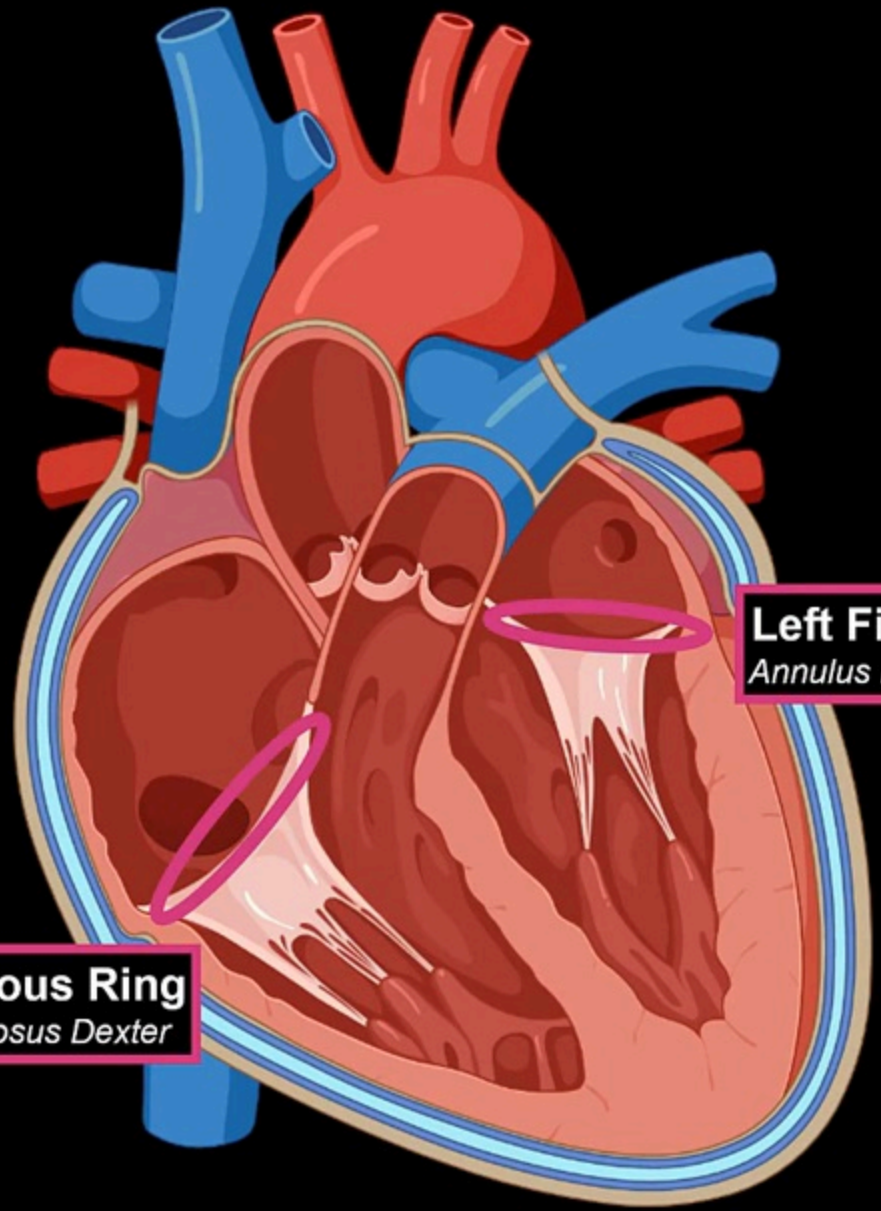




Myocardium



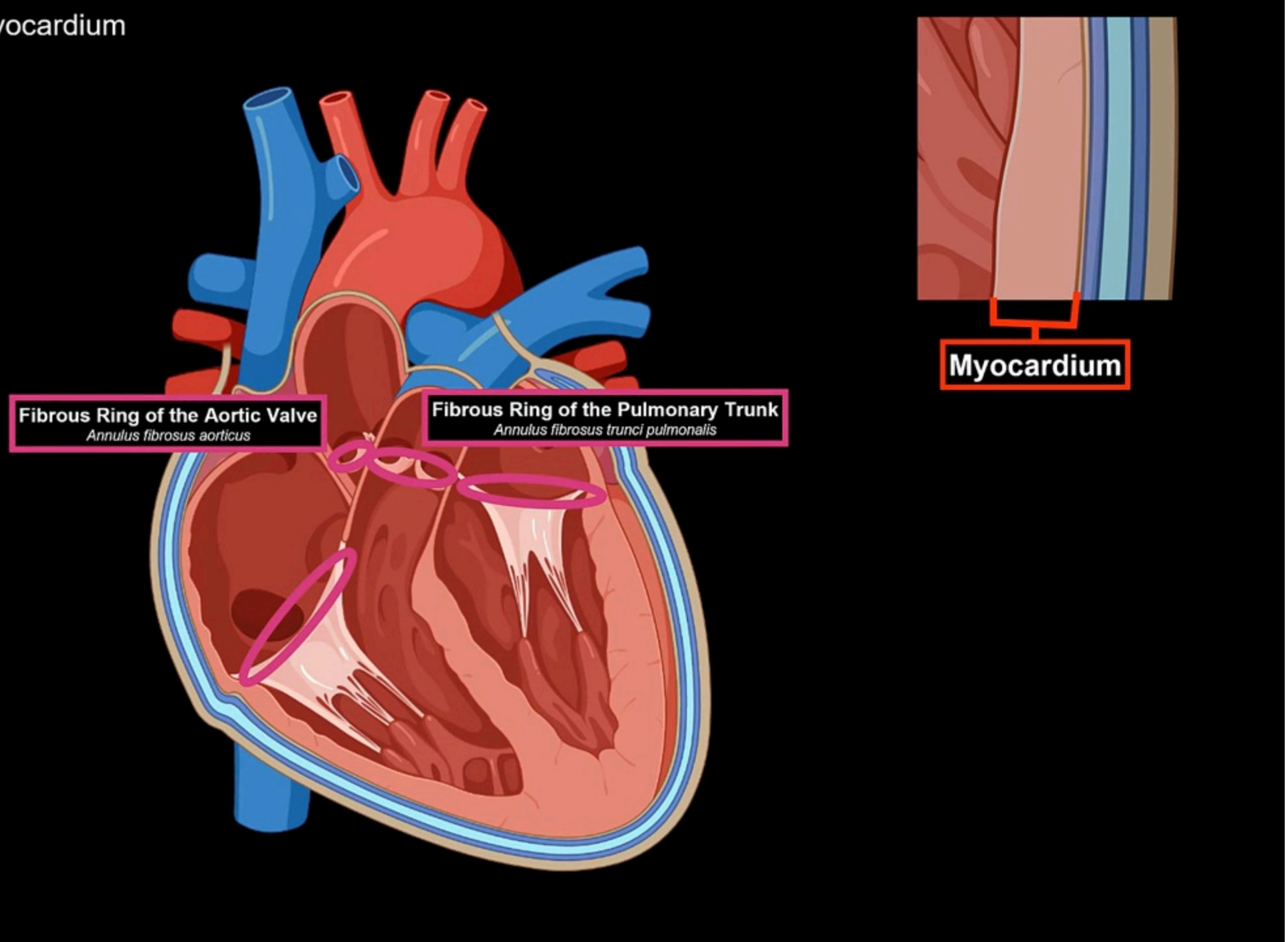
Myocardium



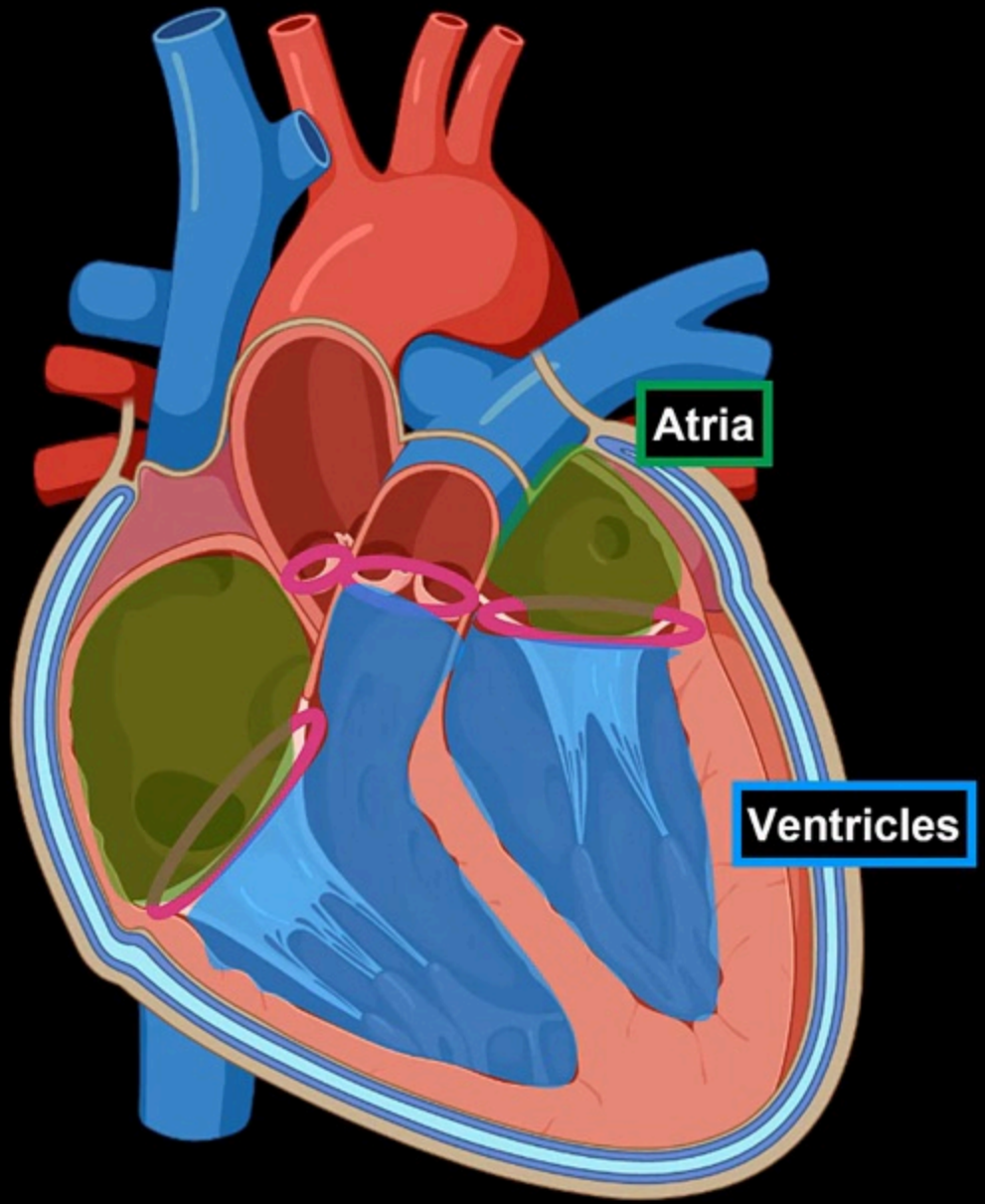
Left Fibrous Ring
Annulus Fibrosus sinister

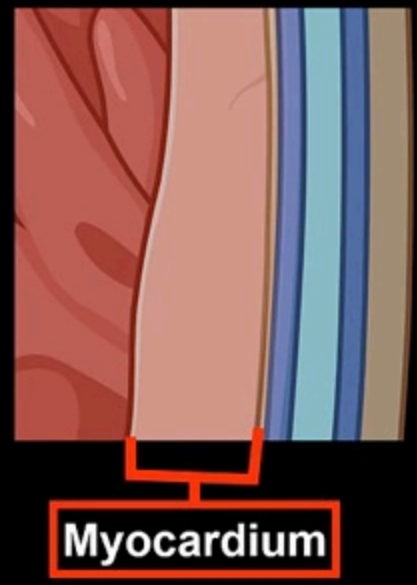
Right Fibrous Ring
Annulus Fibrosus Dexter

Myocardium

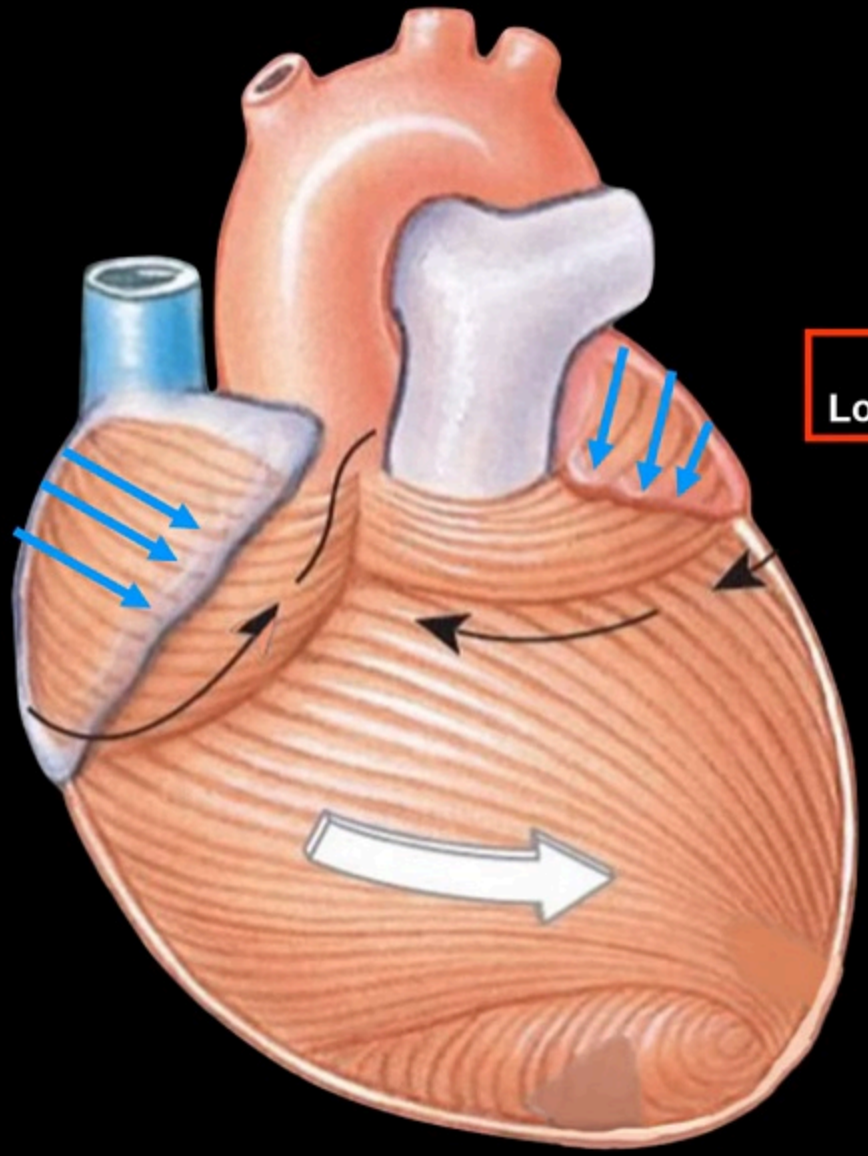

Fibrous Ring of the Aortic Valve
Annulus fibrosus aorticus
Fibrous Ring of the Pulmonary Trunk
Annulus fibrosus trunci pulmonalis
Myocardium

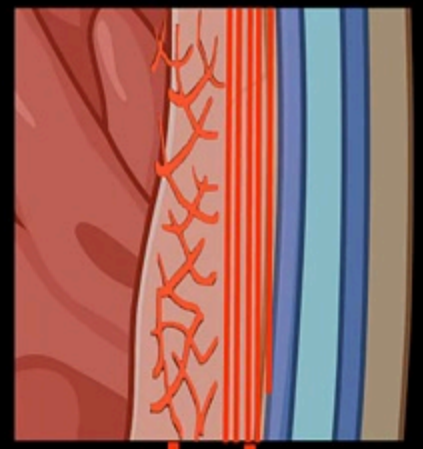
Myocardium





Myocardium of the Atria

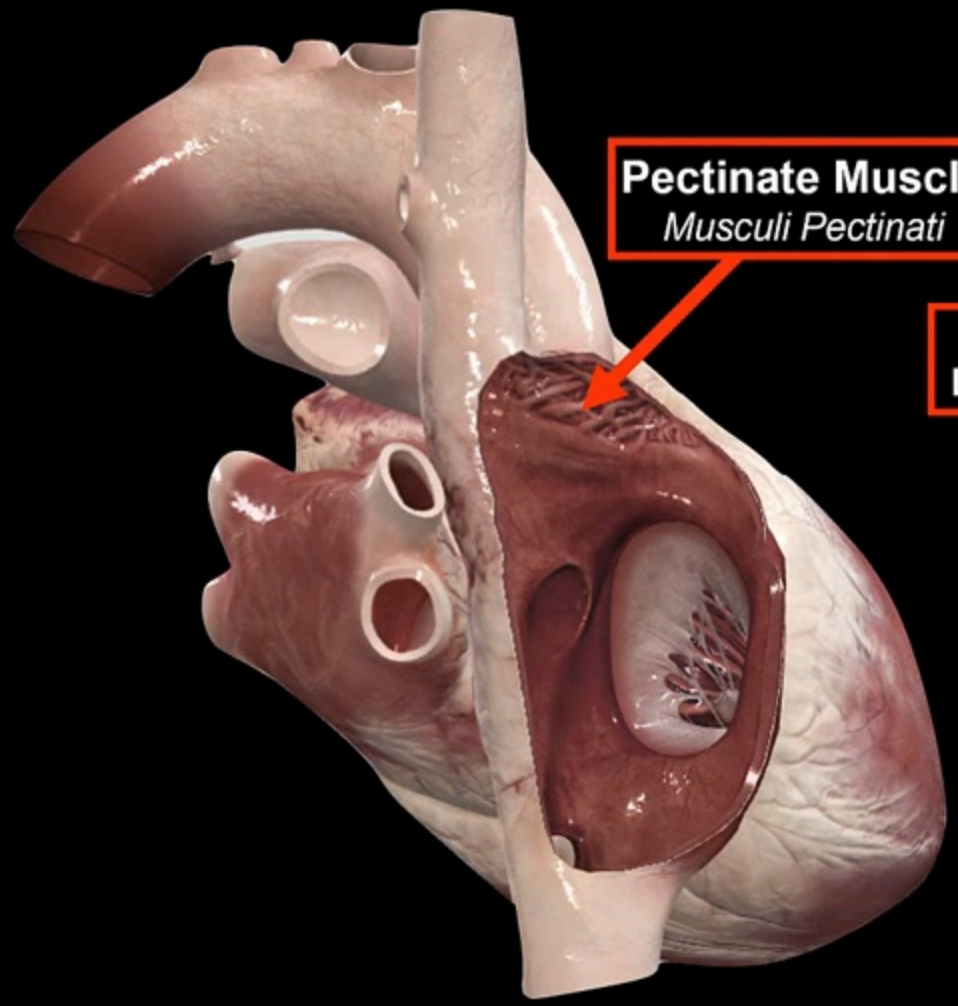




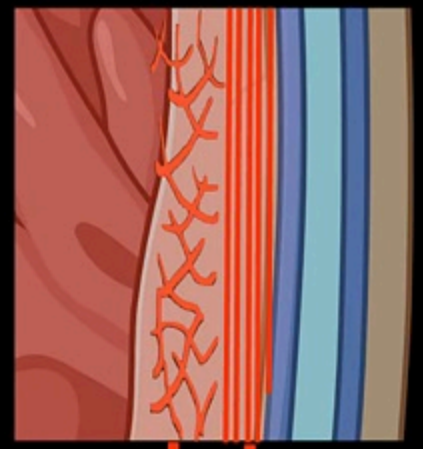
Deep Layer:
Longitudinal Muscle fibers

Superficial Layer:
Circular Muscle fibers

Myocardium of the Atria



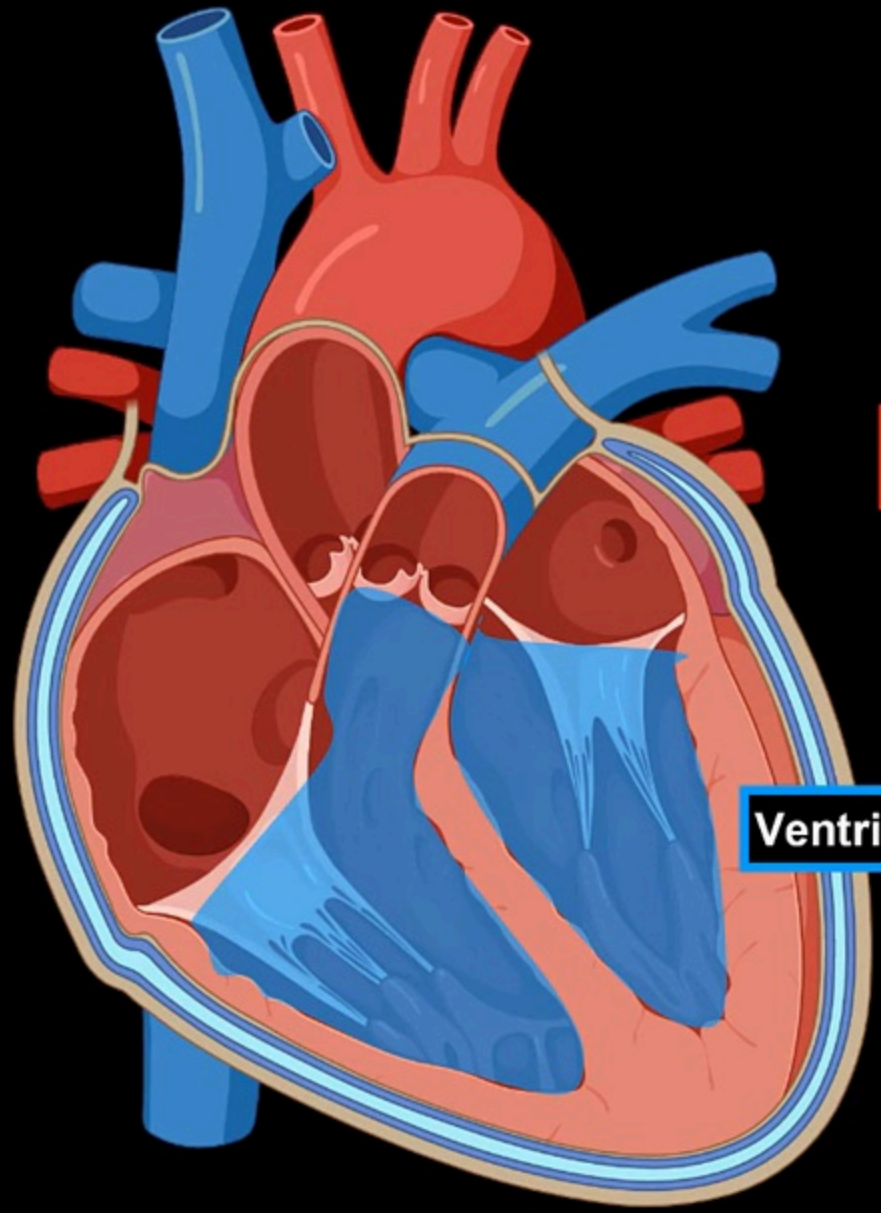
Pectinate Muscle
Musculi Pectinati



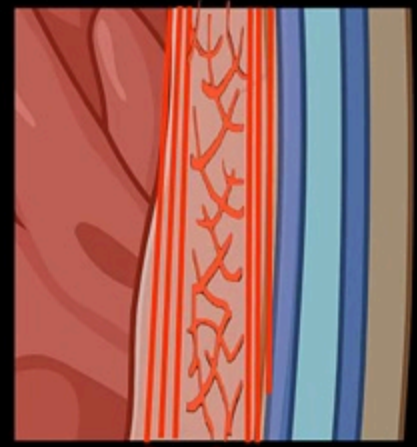
Deep Layer:
Longitudinal Muscle fibers

Superficial Layer:
Circular Muscle fibers

Myocardium of the Ventricles



Ventricles

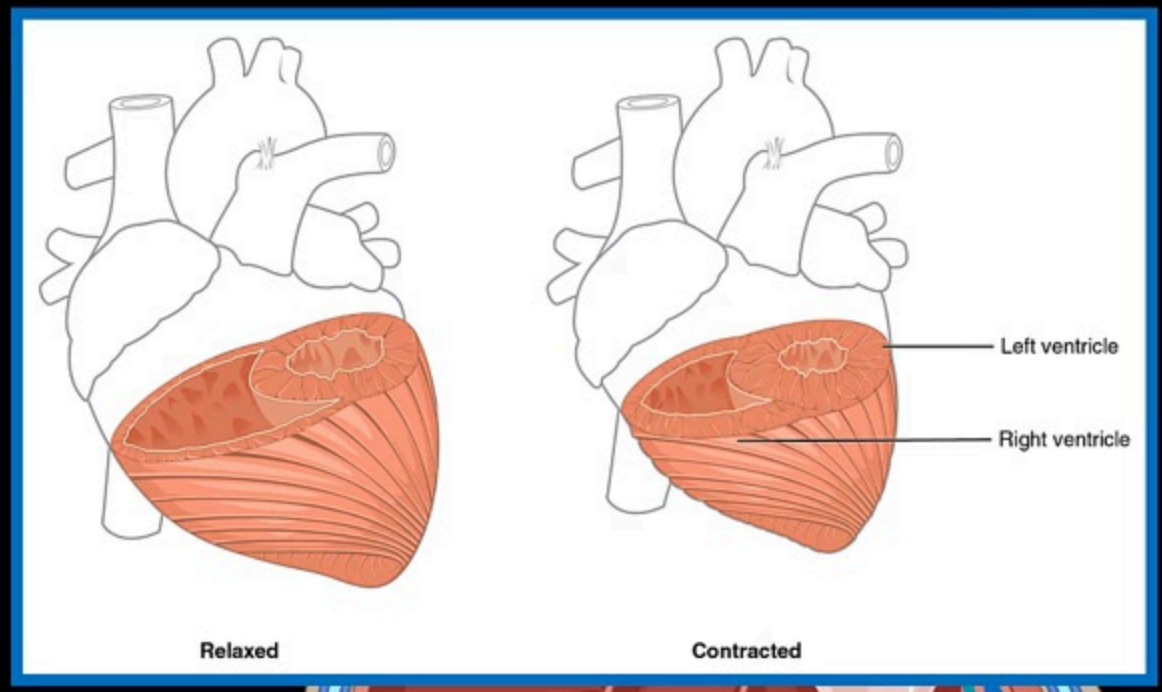


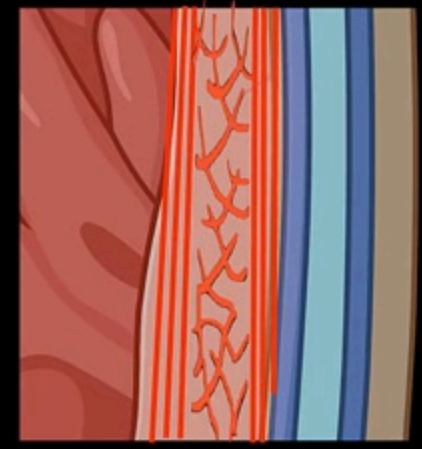
Deep Layer:
Longitudinal Muscle fibers

Middle Layer:
Circular Muscle fibers

Superficial Layer:
Longitudinal Fibers

Myocardium of the Ventricles





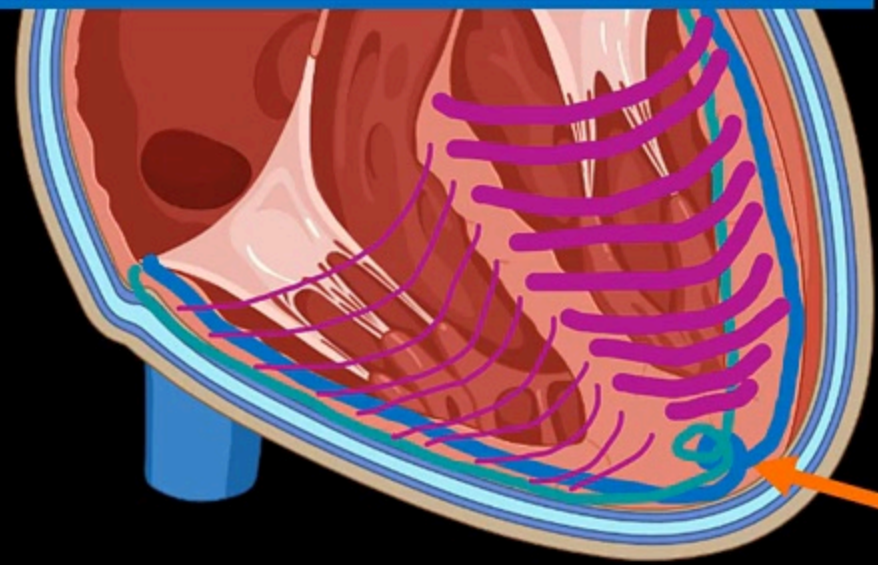
Longitudinal Muscle fibers

Superfical —> Apex —> Deep

Deep —> Apex —> Superficial

Middle Layer:

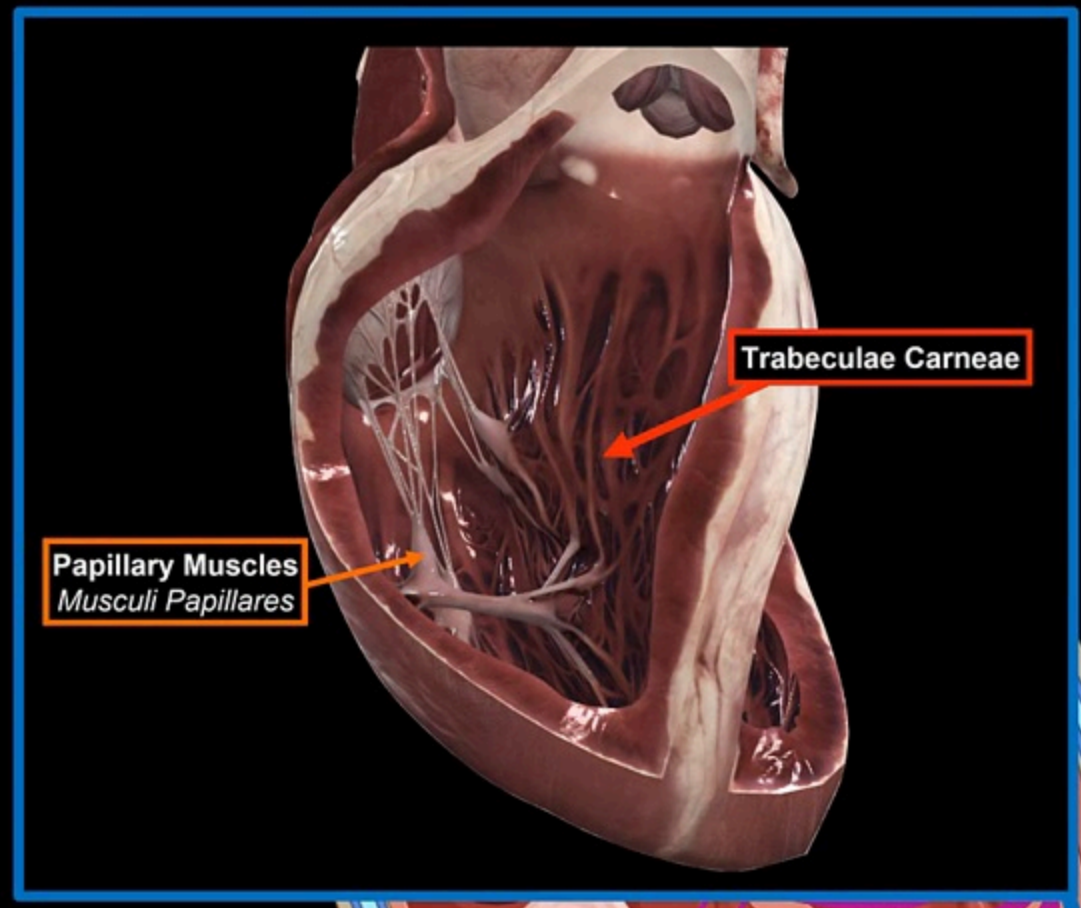
Circular Muscle fibers



Vortex of the Heart

Vortex Cordis

Myocardium of the Ventricles





Longitudinal Muscle fibers

Superfical —> Apex —> Deep

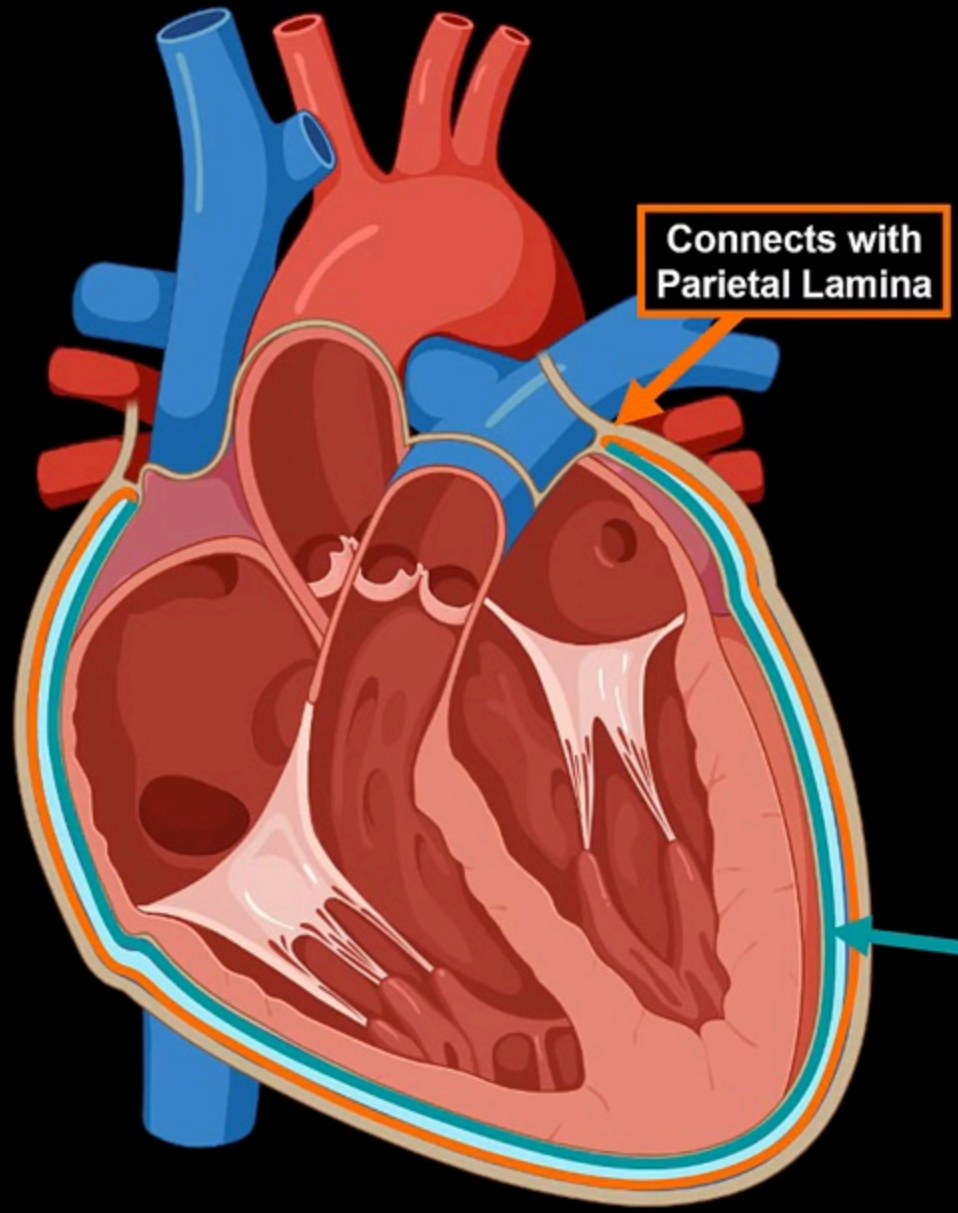
Deep —> Apex —> Superficial

Middle Layer:

Circular Muscle fibers

Vortex of the Heart
Vortex Cordis

Epicardium

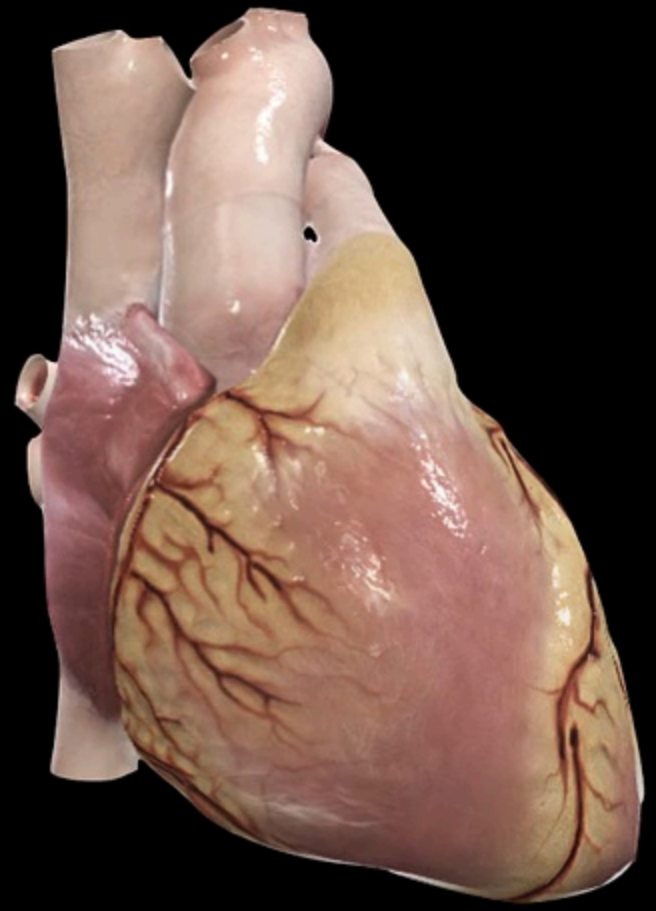




Epicardium

Visceral Lamina of
Serous Pericardium

Pericardium





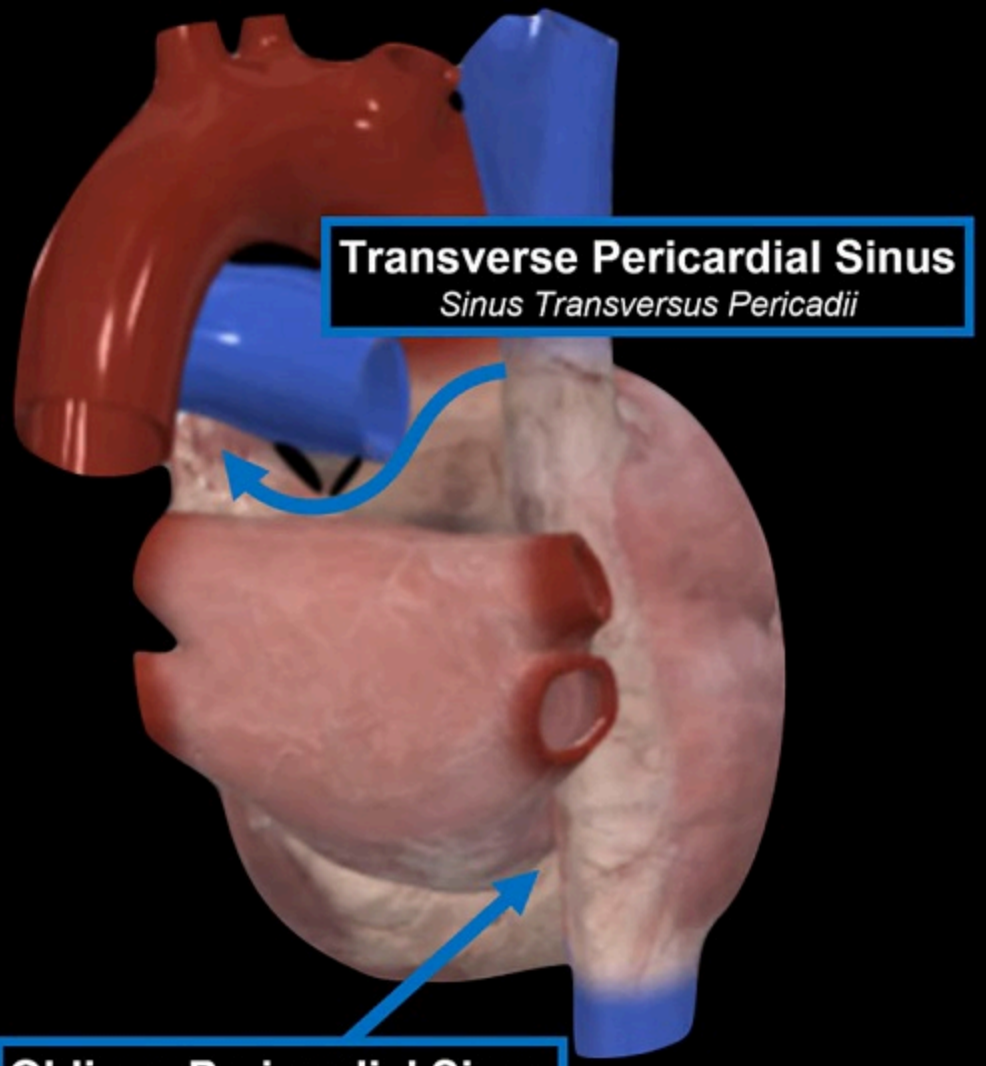
Visceral lamina (Epicardium)

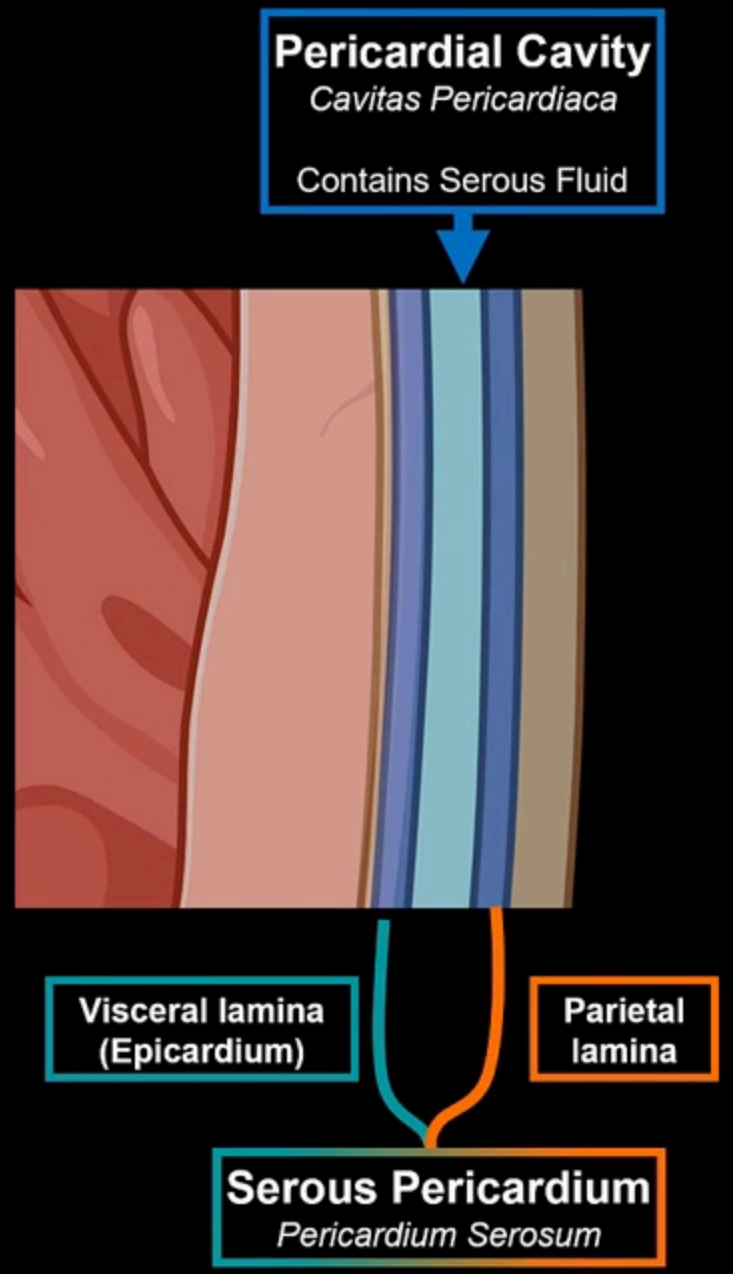
Parietal lamina

Serous Pericardium

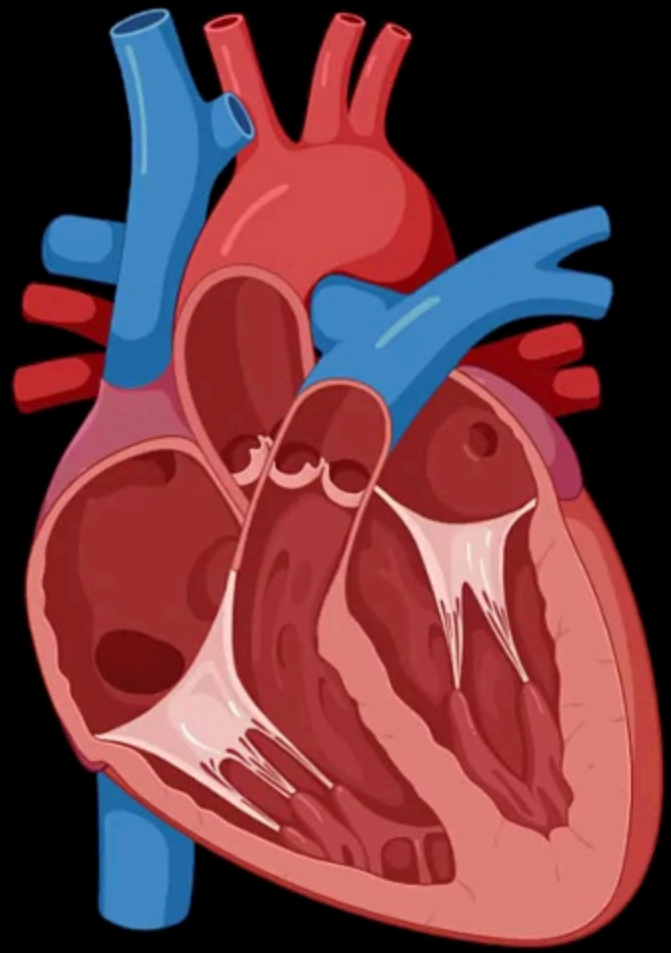
Pericardium Serosum

Pericardium





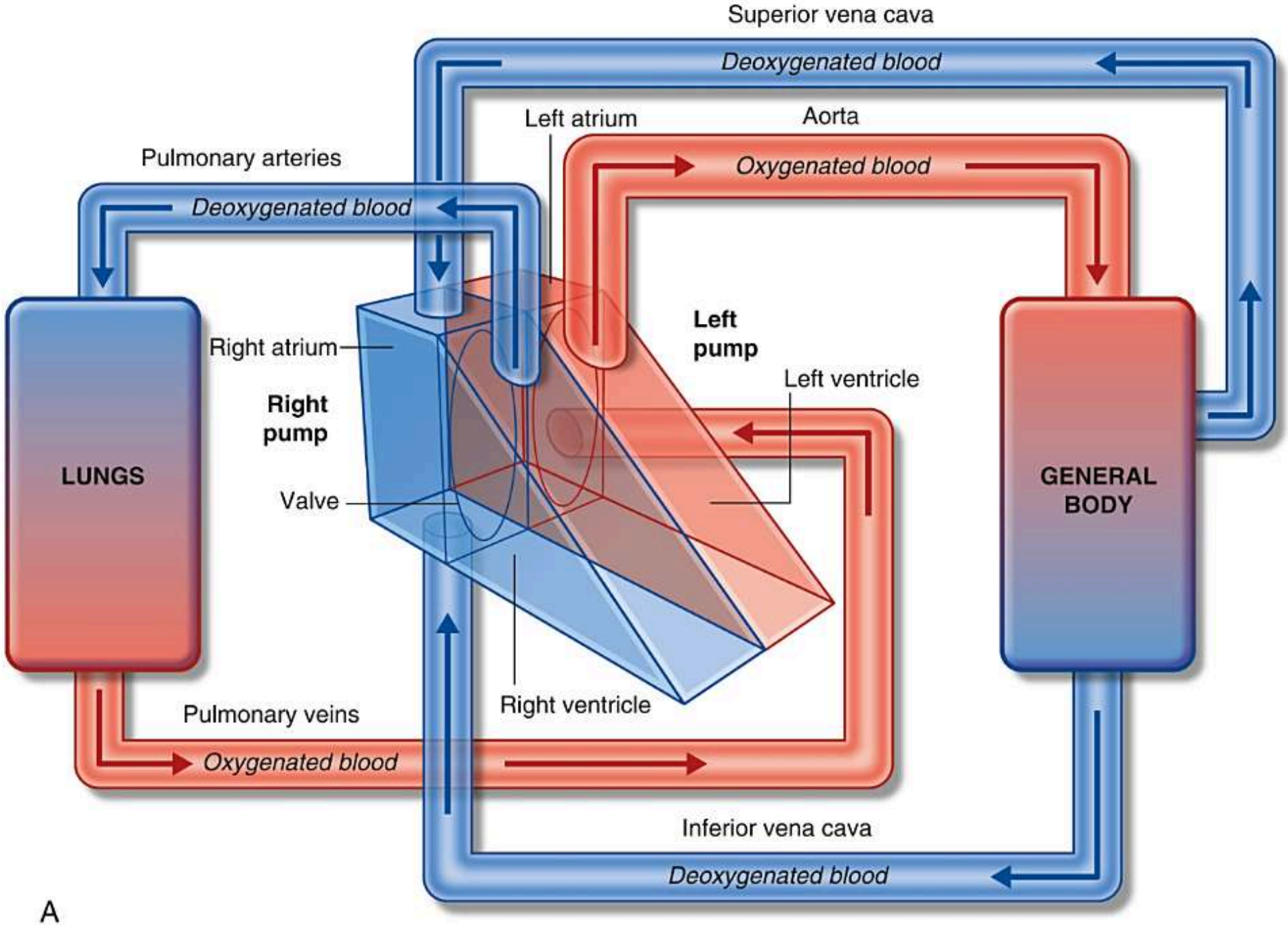
Heart Anatomy



Internal Features of the Heart

- ✓ Right atrium
- ✓ Left atrium
- ✓ Left ventricle
- ✓ Right ventricle
- ✓ Interventricular septum




Superior vena cava
Deoxygenated blood
Left atrium
Aorta
Pulmonary arteries
Oxygenated blood
Deoxygenated blood
Left pump
Right atrium
Left ventricle
Right pump
LUNGS
GENERAL BODY
Valve
Right ventricle
Pulmonary veins
Oxygenated blood
Inferior vena cava
Deoxygenated blood

A

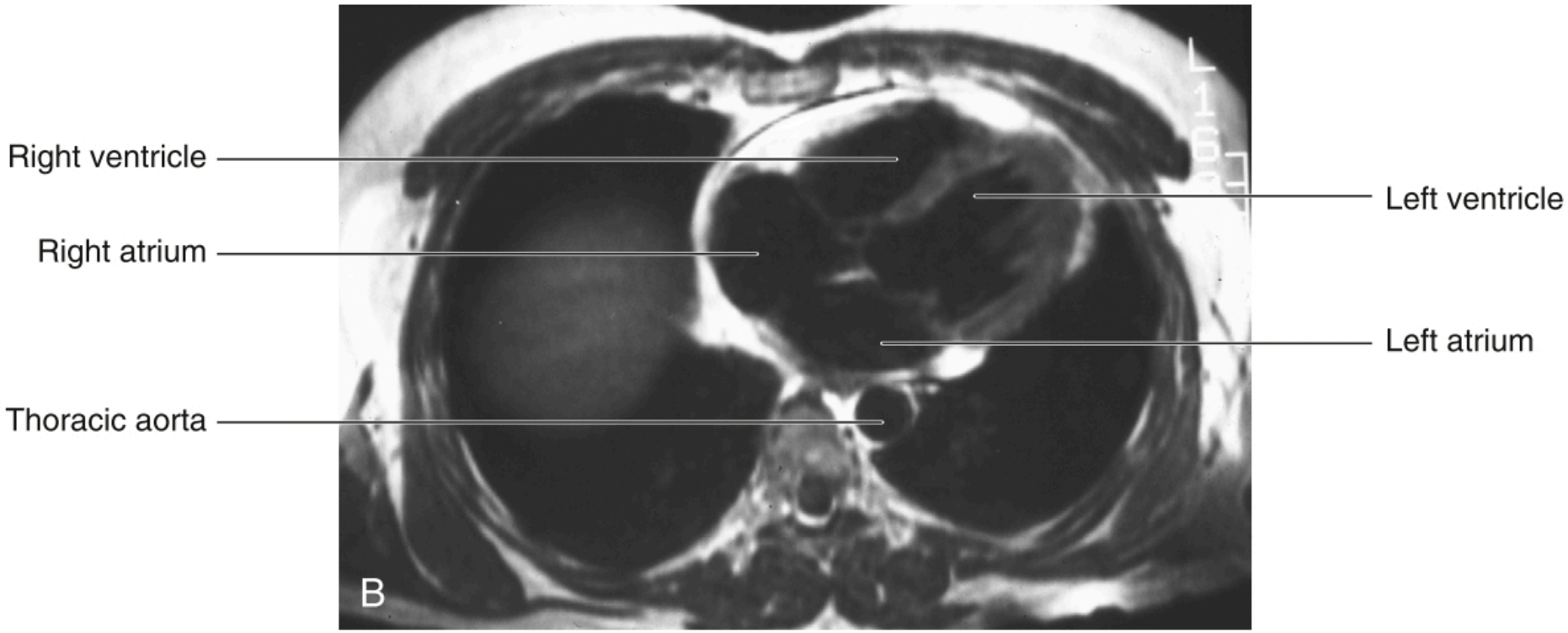


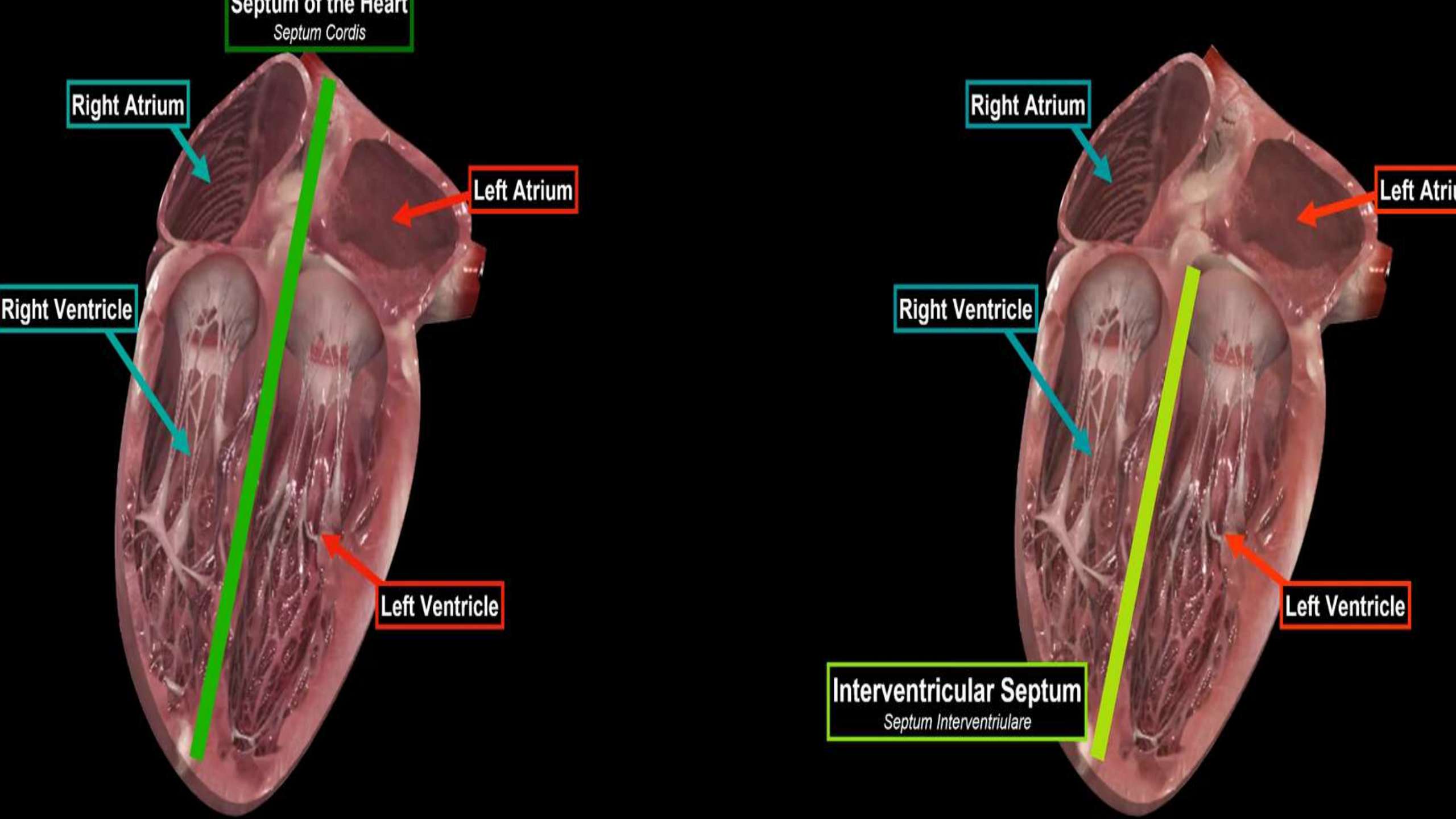
Fig. 3.71 (**A**) The heart has two pumps. (**B**) Magnetic resonance image of midthorax showing all four chambers and septa.

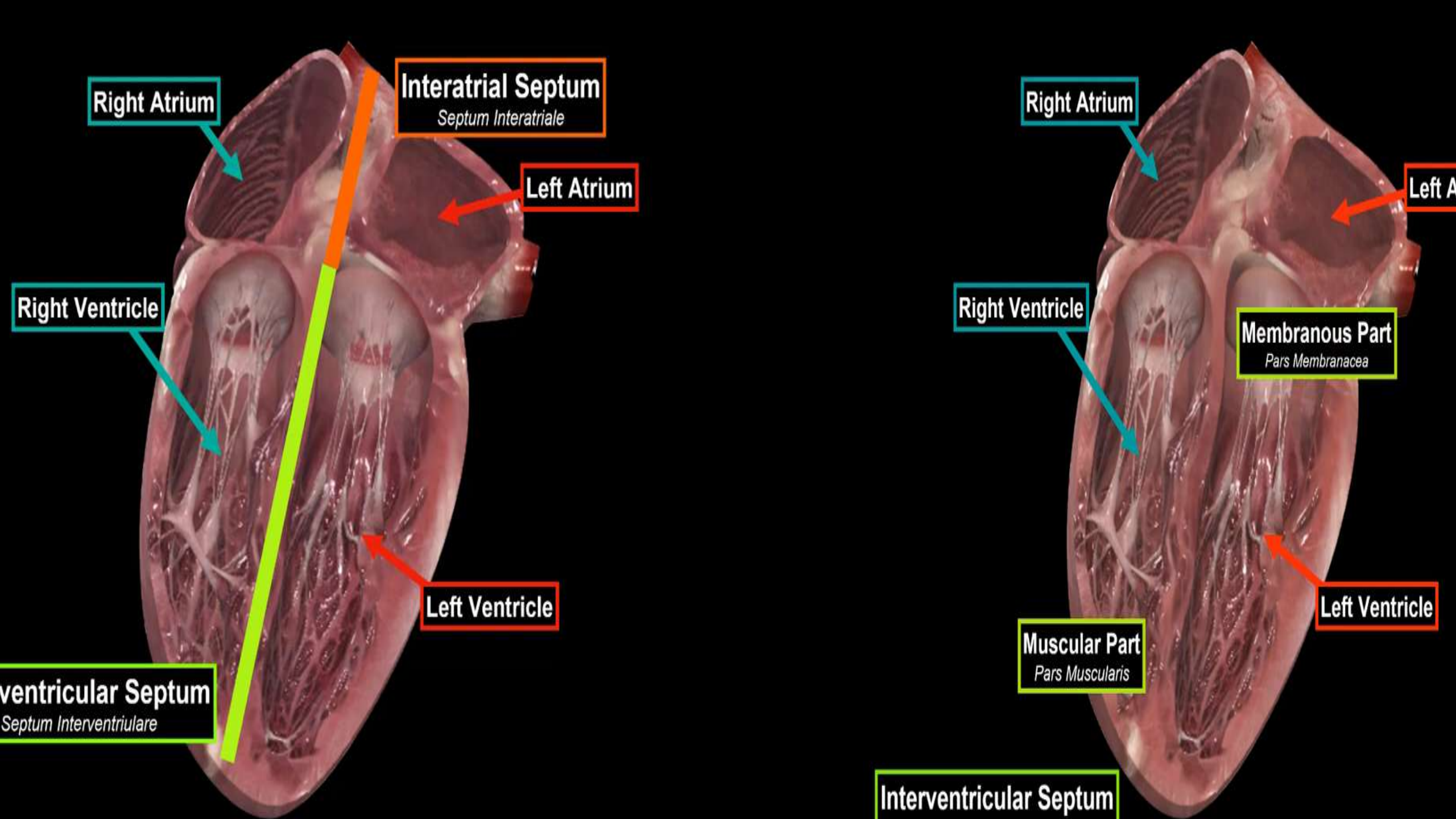
❖ Rules :



1. Each chamber of the heart is formed of:
   a. Smooth part
   b. Rough part

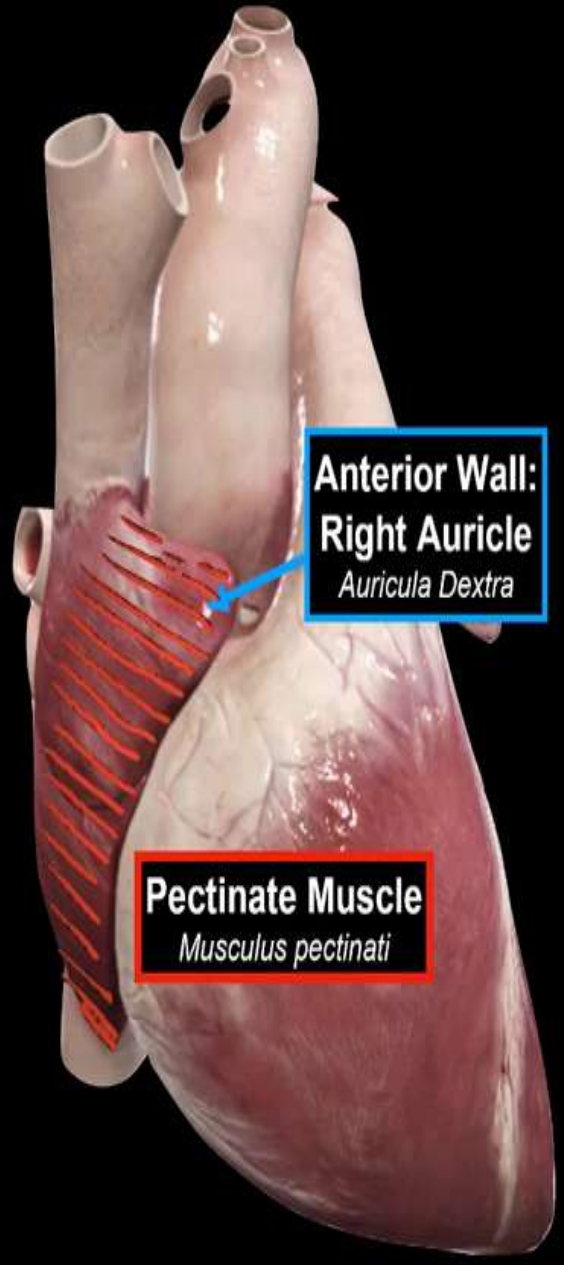
2. The smooth part of each chamber is developed by absorption from the nearby chamber


Septum of the Heart
Septum Cordis
Right Atrium
Left Atrium
Right Ventricle
Left Ventricle
Right Atrium
Left Atri
Right Ventricle
Left Ventricle
Interventricular Septum
Septum Interventriulare


Right Atrium
Interatrial Septum
Septum Interatriale
Left Atrium
Right Ventricle
Left Ventricle
ventricular Septum
Septum Interventriulare
Right Atrium
Left A
Right Ventricle
Membranous Part
Pars Membranacea
Left Ventricle
Muscular Part
Pars Muscularis
Interventricular Septum



Lateral Wall



Right Atrium

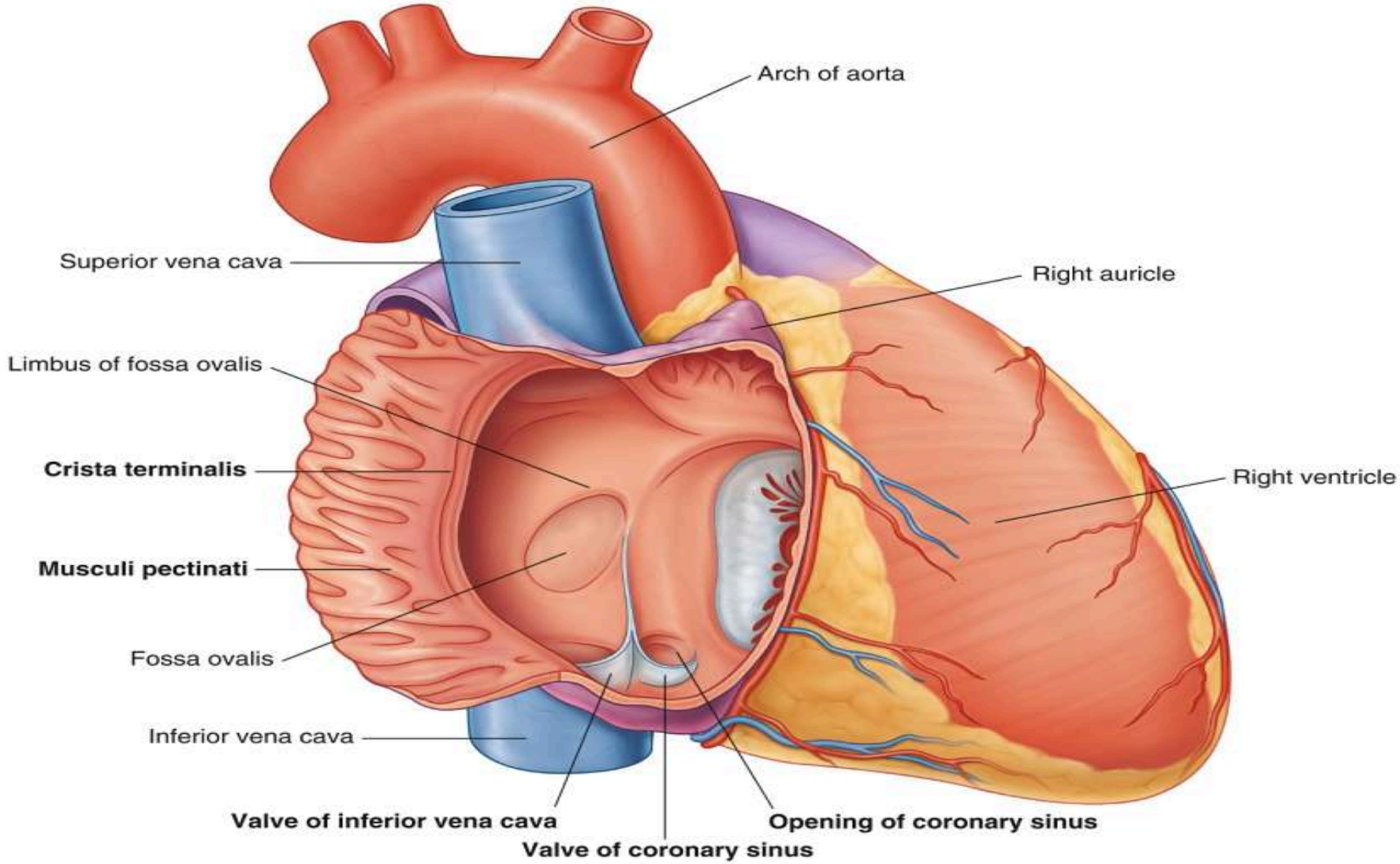
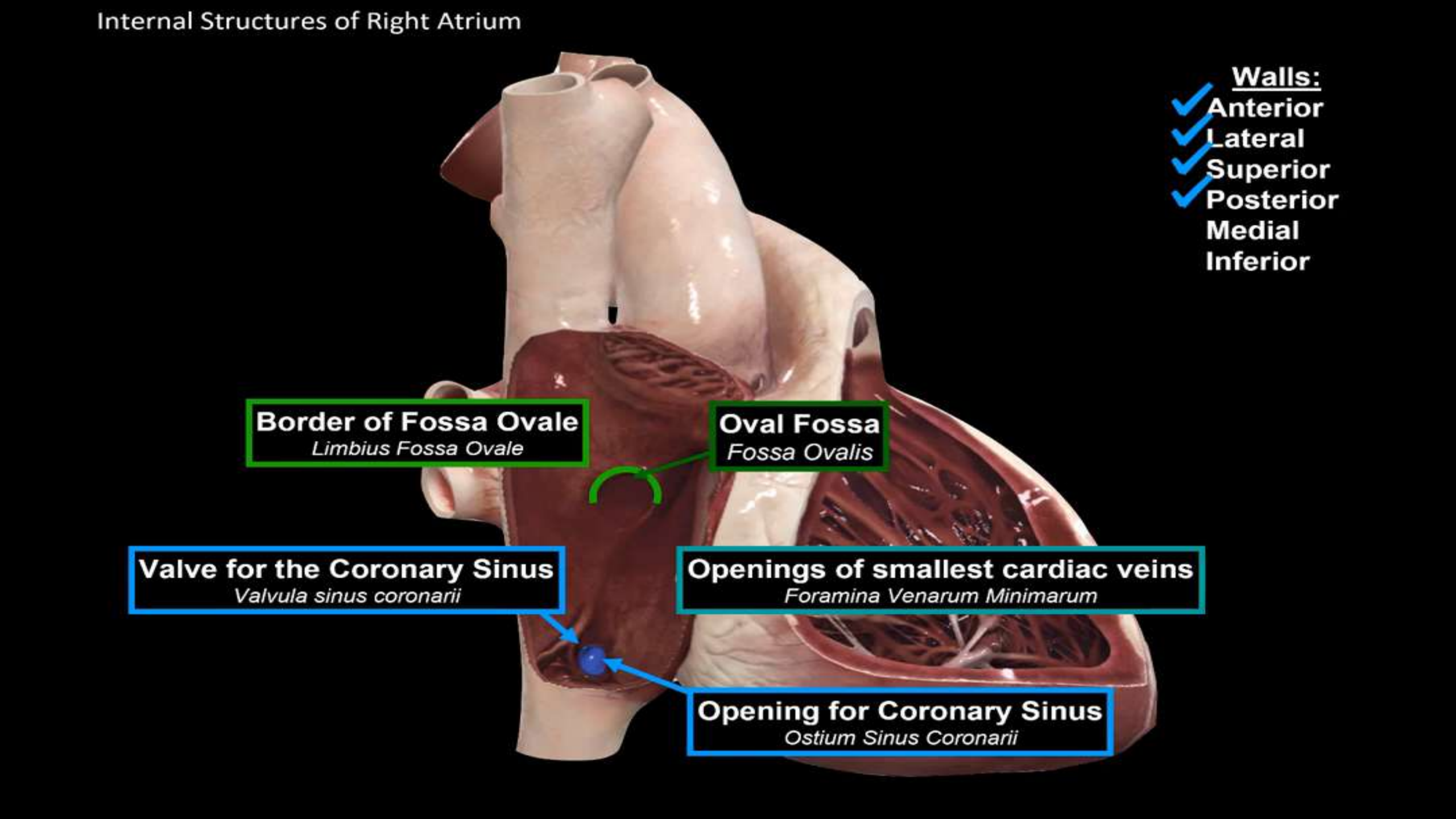


Fig. 3.72 Internal View of Right Atrium.

Internal Structures of Right Atrium



Chambers of the heart

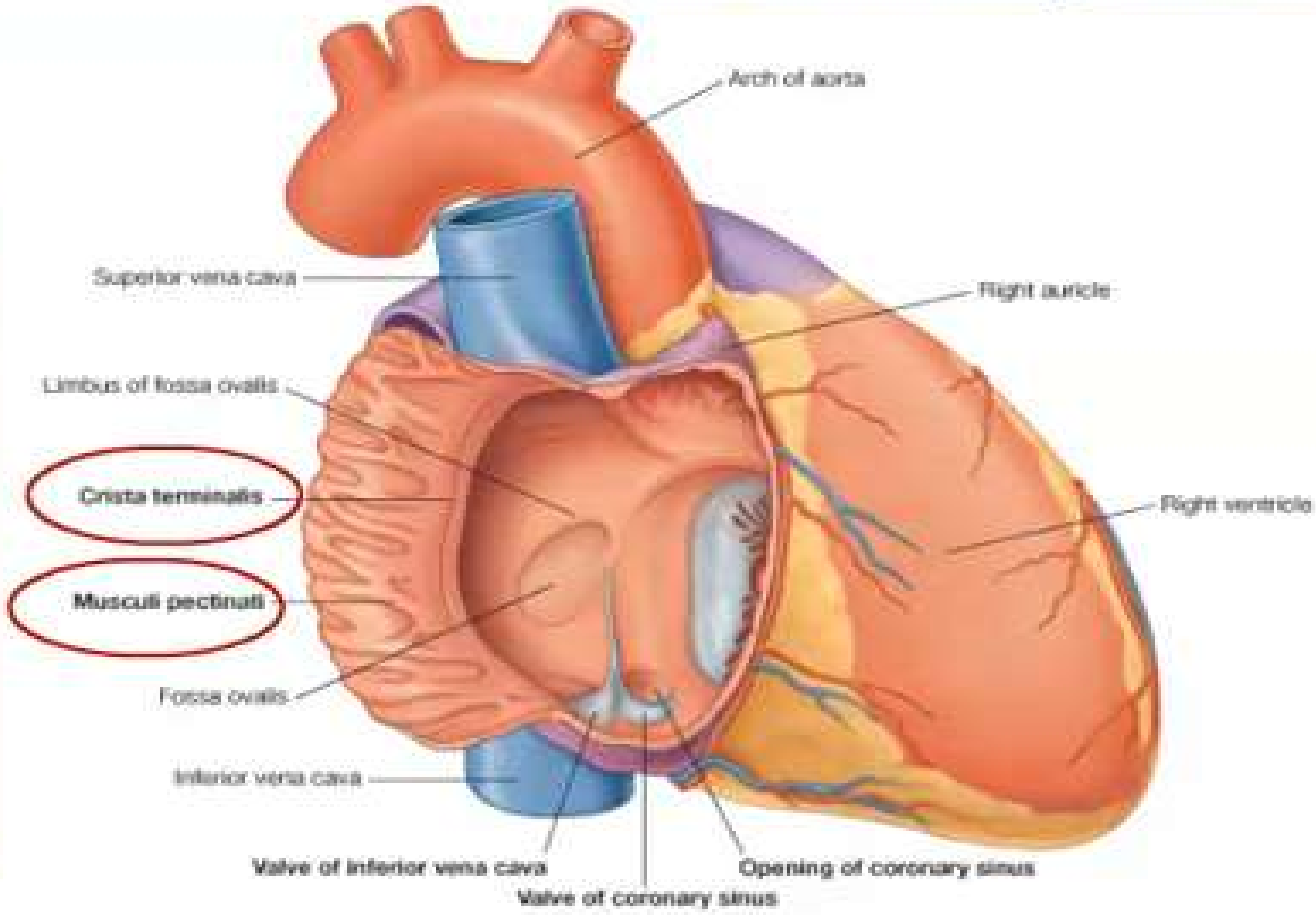
Right atrium: it forms.....of the heart?

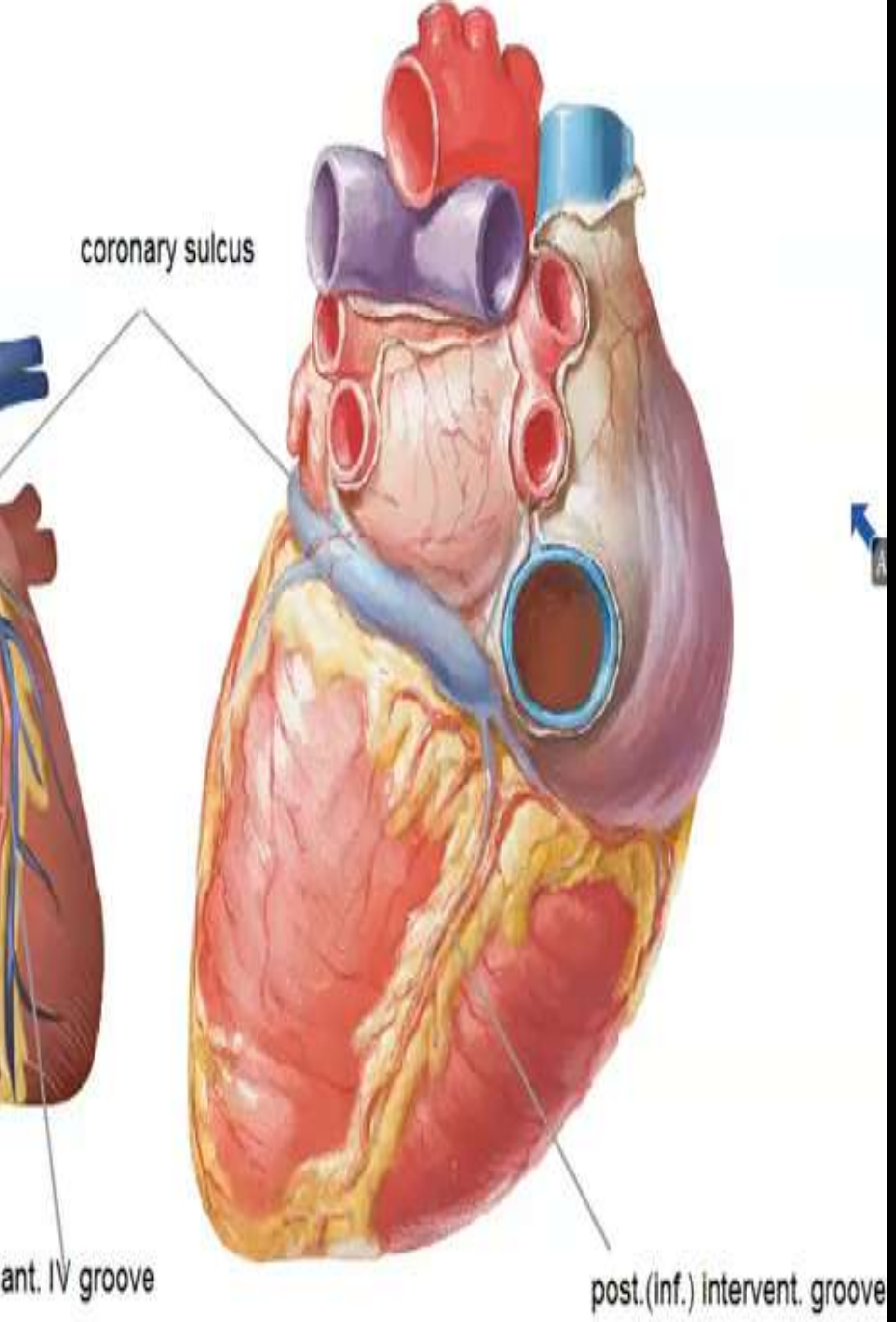
Interior of right atrium

Crista terminalis corresponds to sulcus terminalis on the external surface of the right atrium.

Rough part

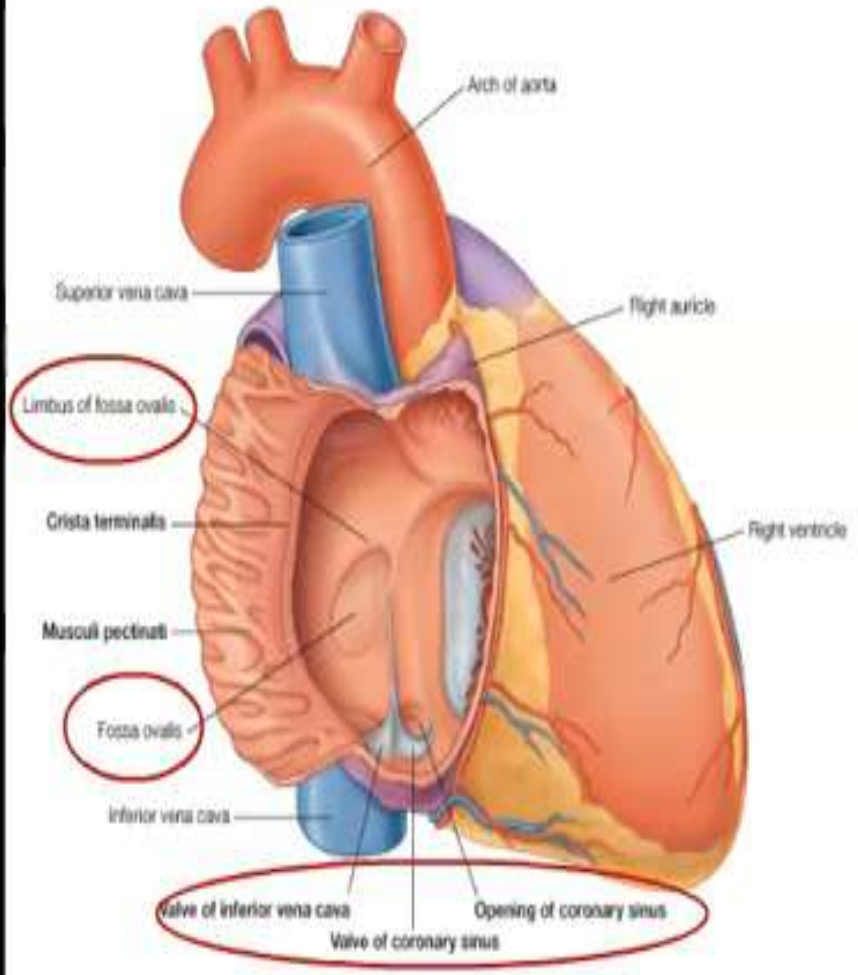
- It is the internal aspect of the anterior wall.
- It has pectinate muscles (parallel muscular ridges) which extends to the interior of right auricle.
- These pectinate muscles are branching from longitudinal muscular ridge called (crista terminalis).





Interior of right atrium

Rough part

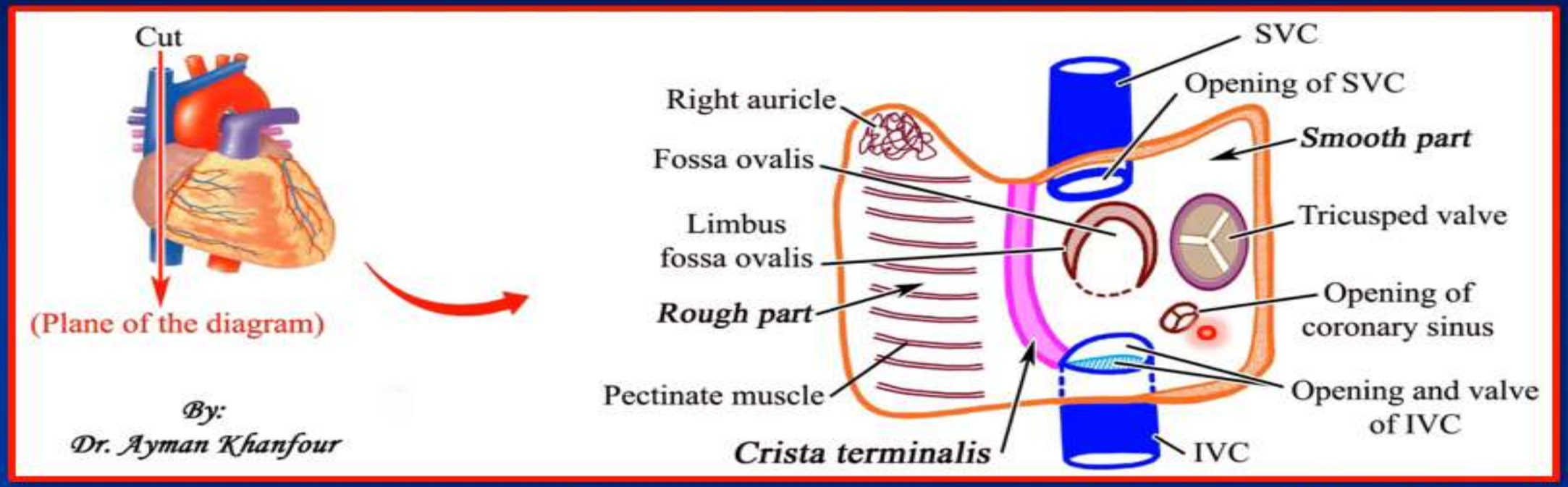


Smoo

- It is the internal aspect of wall.
- It receives the openings of (without valves), IVC (with valve), and coronary sinus (developed valve).
- There is also the tricuspid right atrioventricular orifice guarded by tricuspid valve.

Notice fossa ovalis and its septal wall. See embryology

Right Atrium



Parts

- The Rt. Atrium is divided by crista terminalis into:
  1. Smooth part
  2. Rough part (Pectinate m.)

Structures

1. Fossa ovalis
2. Limbous fossa ovalis

Openings

1. Tricusped valve
2. SVC
3. IVC
4. Anterior cardiac v.
5. Coronary sinus.
6. Foramina venarum.

Left Atrium

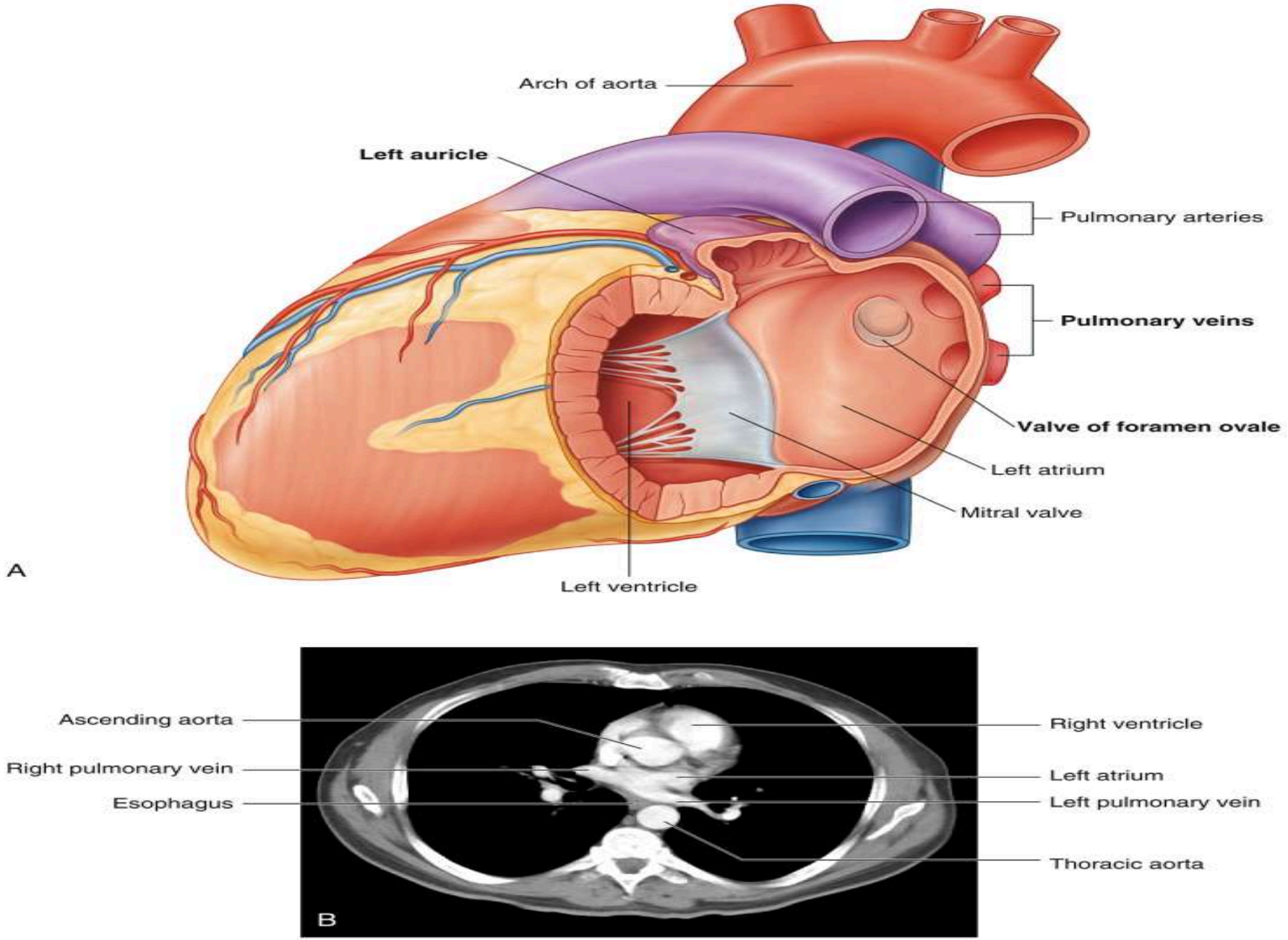
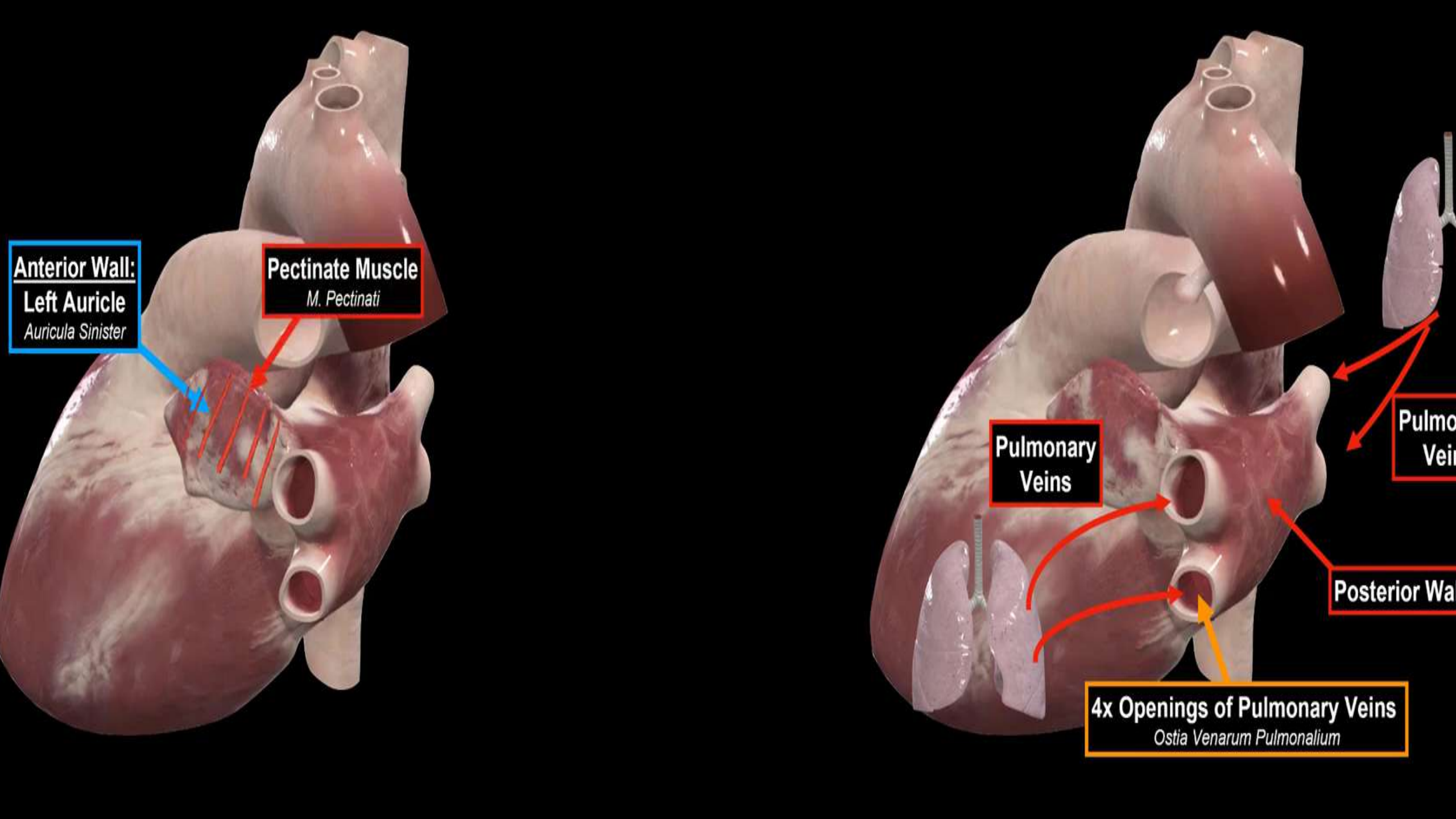
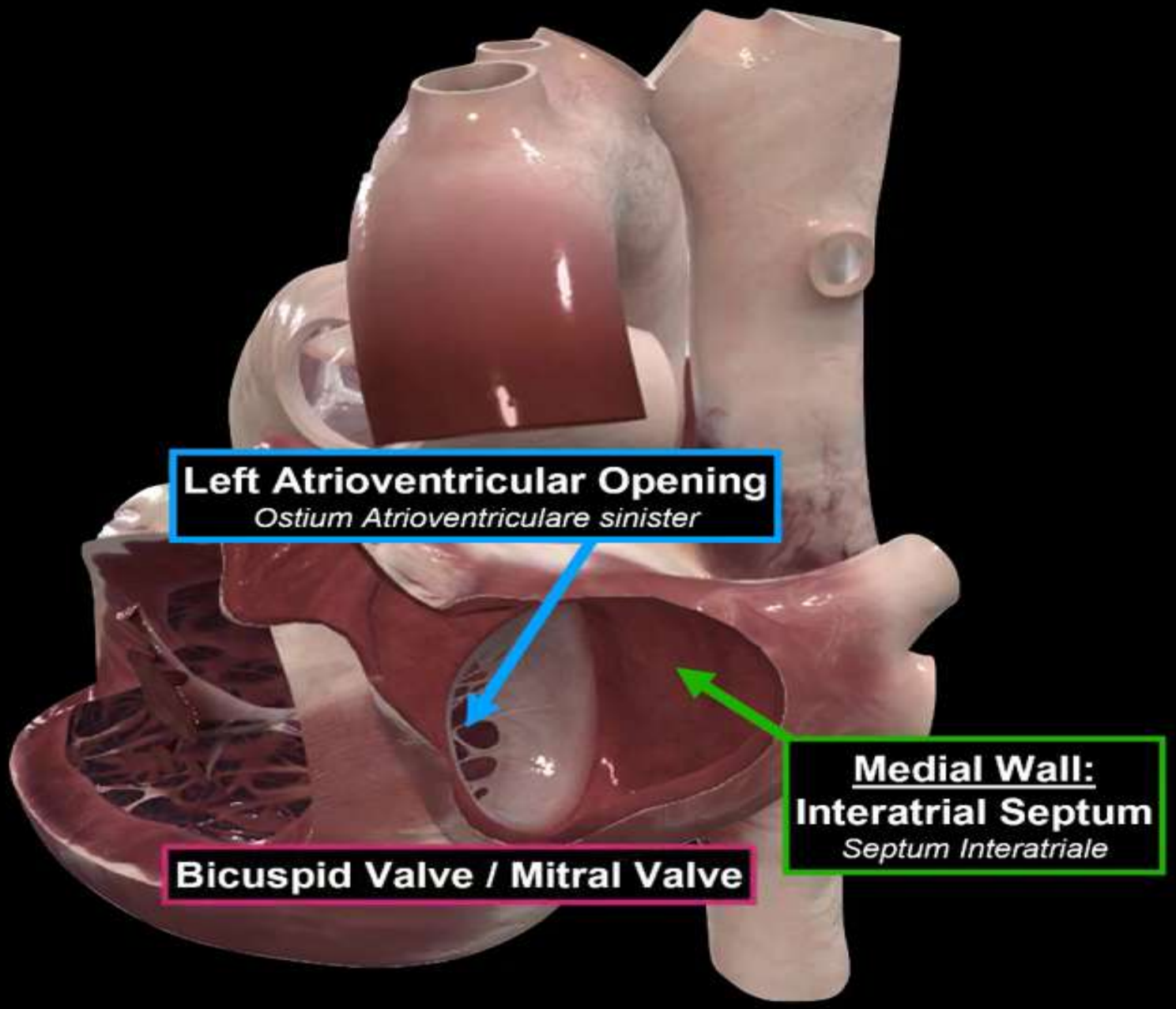


Fig. 3.75 Left Atrium. (**A**) Internal view. (**B**) Axial CT image showing the pulmonary veins entering the left atrium.


Anterior Wall:
Left Auricle
Auricula Sinister
Pectinate Muscle
M. Pectinati
Pulmonary Veins
Pulmo
Vei
Posterior Wa
4x Openings of Pulmonary Veins
Ostia Venarum Pulmonalium

Internal Structures of Left Atrium

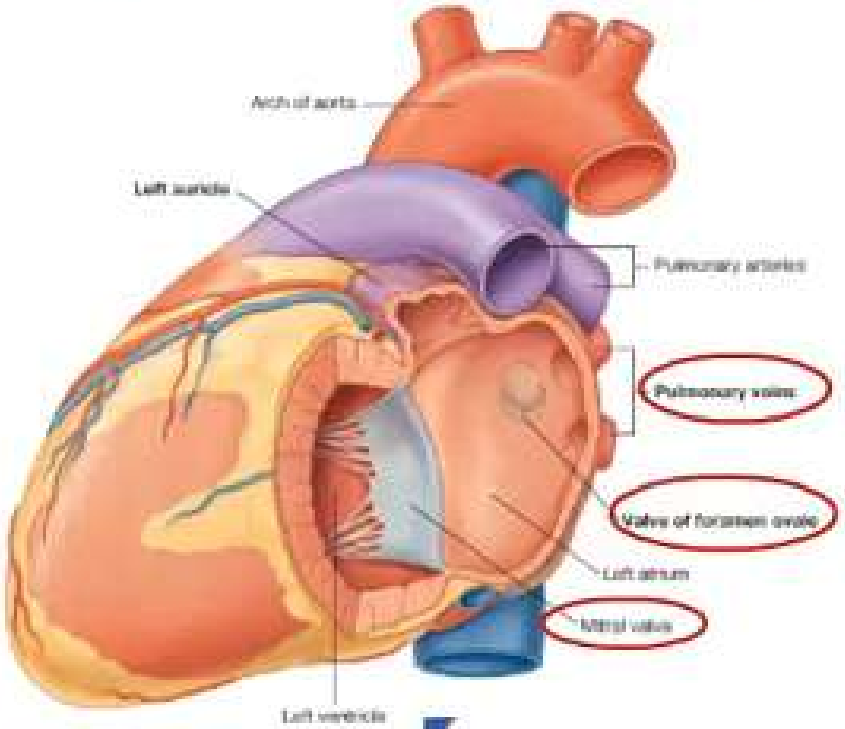


Left atrium: it forms.....of the heart?

Interior of left atrium

Rough part

- Pectinate muscles are present only in the left auricle.

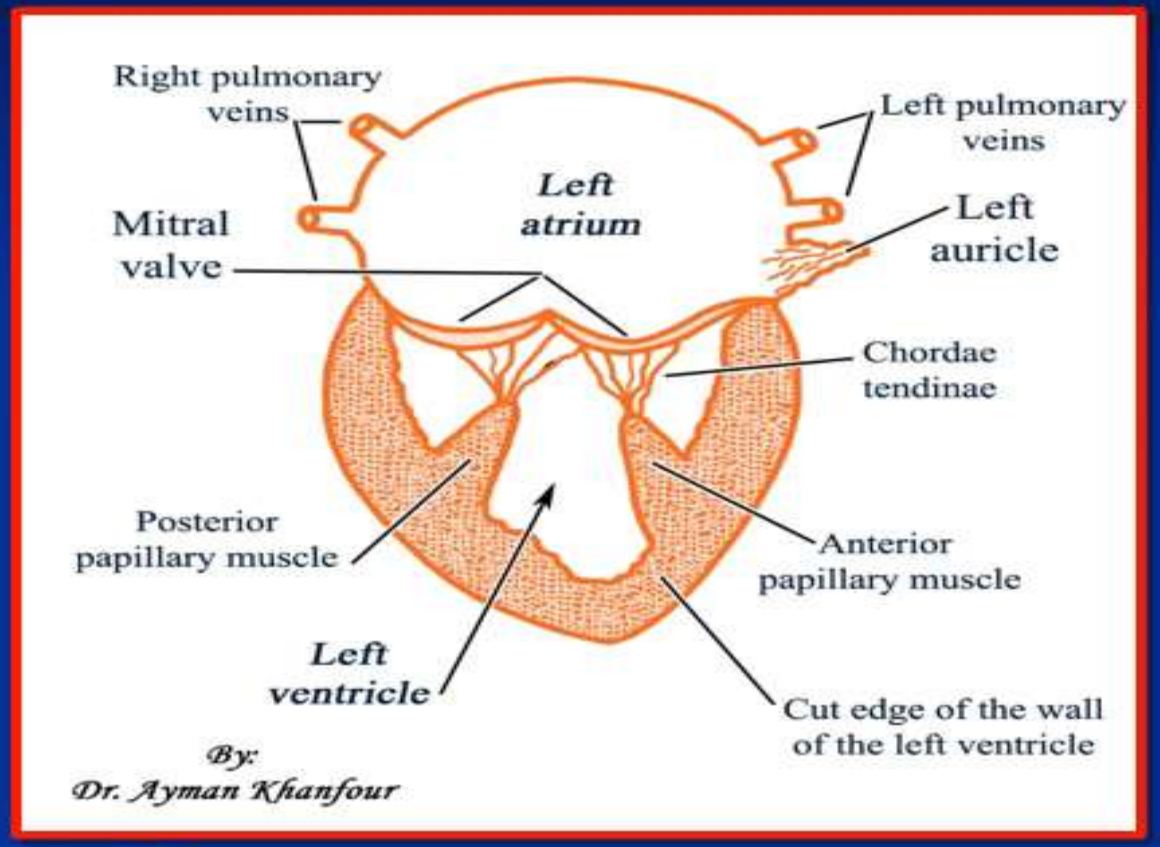


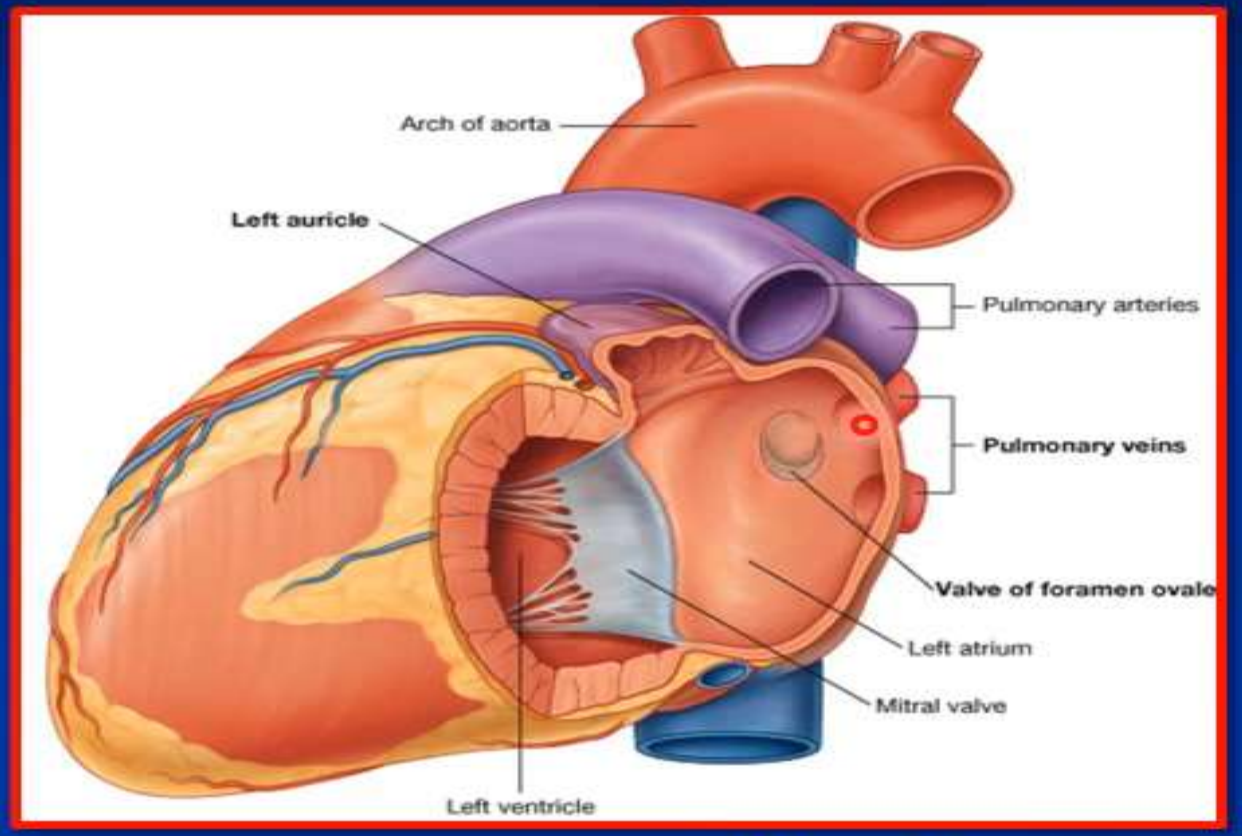
Smooth part

- It forms all the interior aspect of the left atrium (except the auricle).
- It receives the 4 pulmonary veins (all are valveless).
- There is the mitral orifice (left atrioventricular orifice) which is guarded by mitral valve.

- The septal wall shows impression corresponding to fossa ovalis of the right atrium.

Left Atrium





Parts

1. Smooth part
2. Rough part (Lt. auricle)
   (Pectinate m.)

Openings

1. Mitral valve
2. Pulmonary veins

Right Ventricle

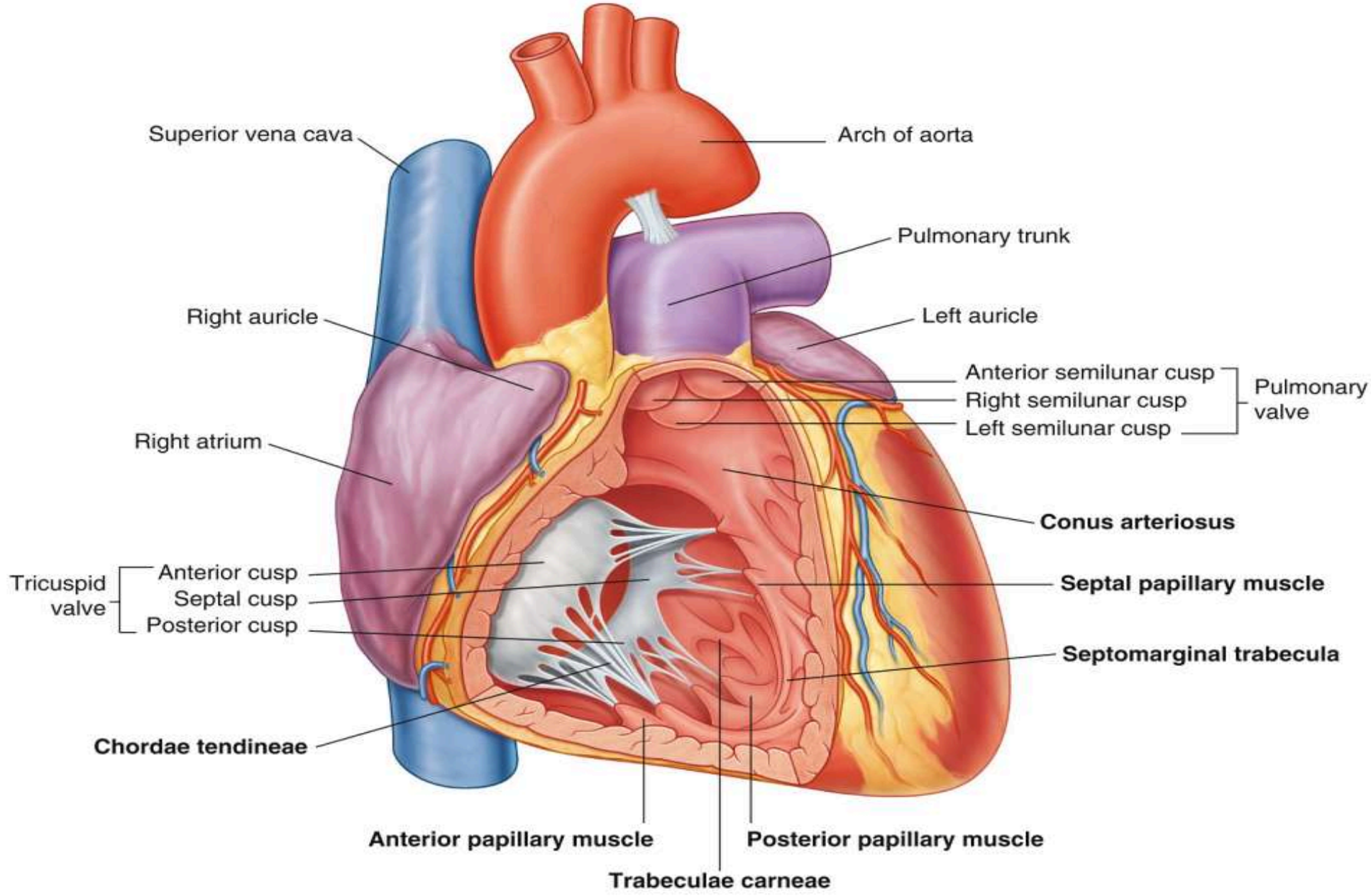
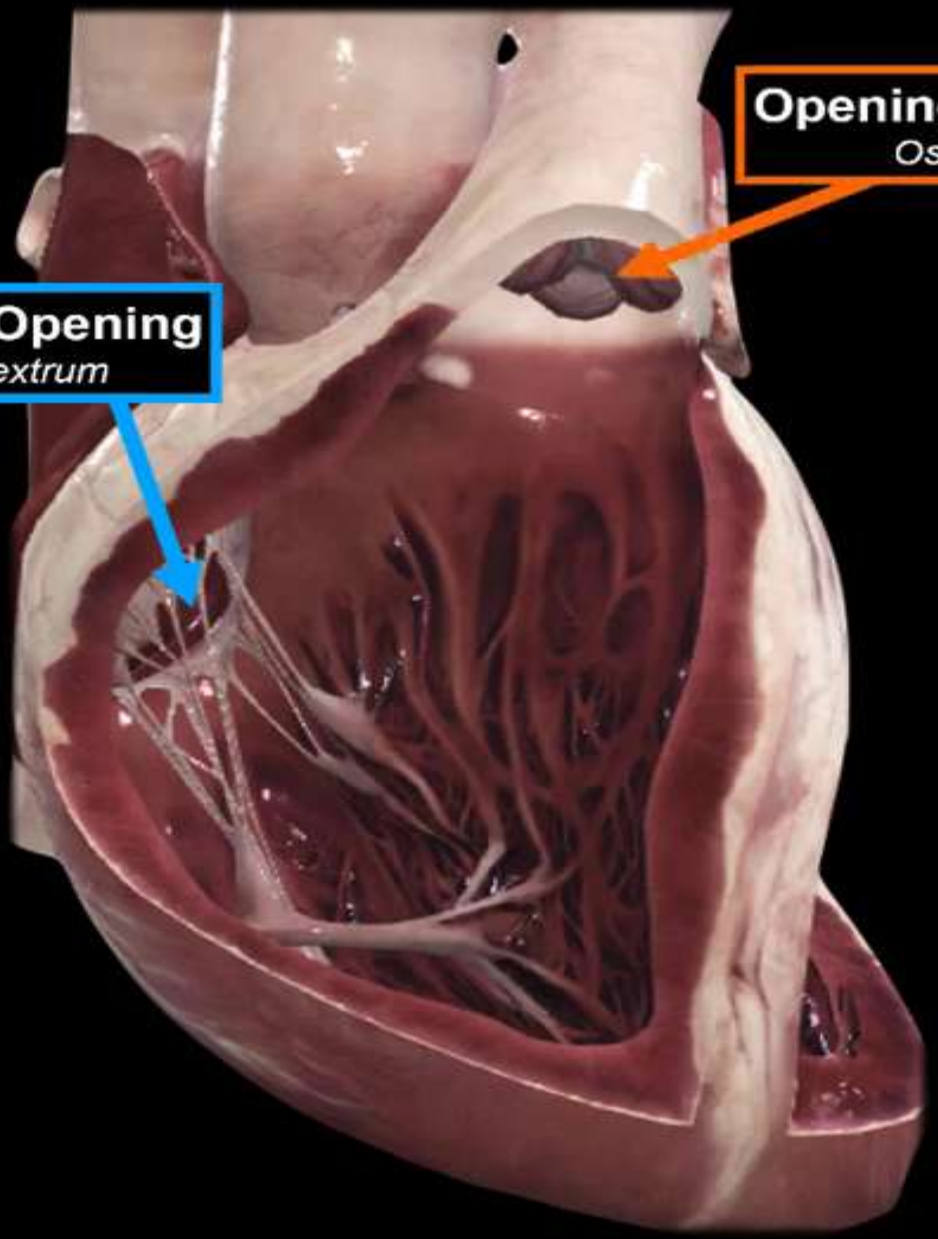
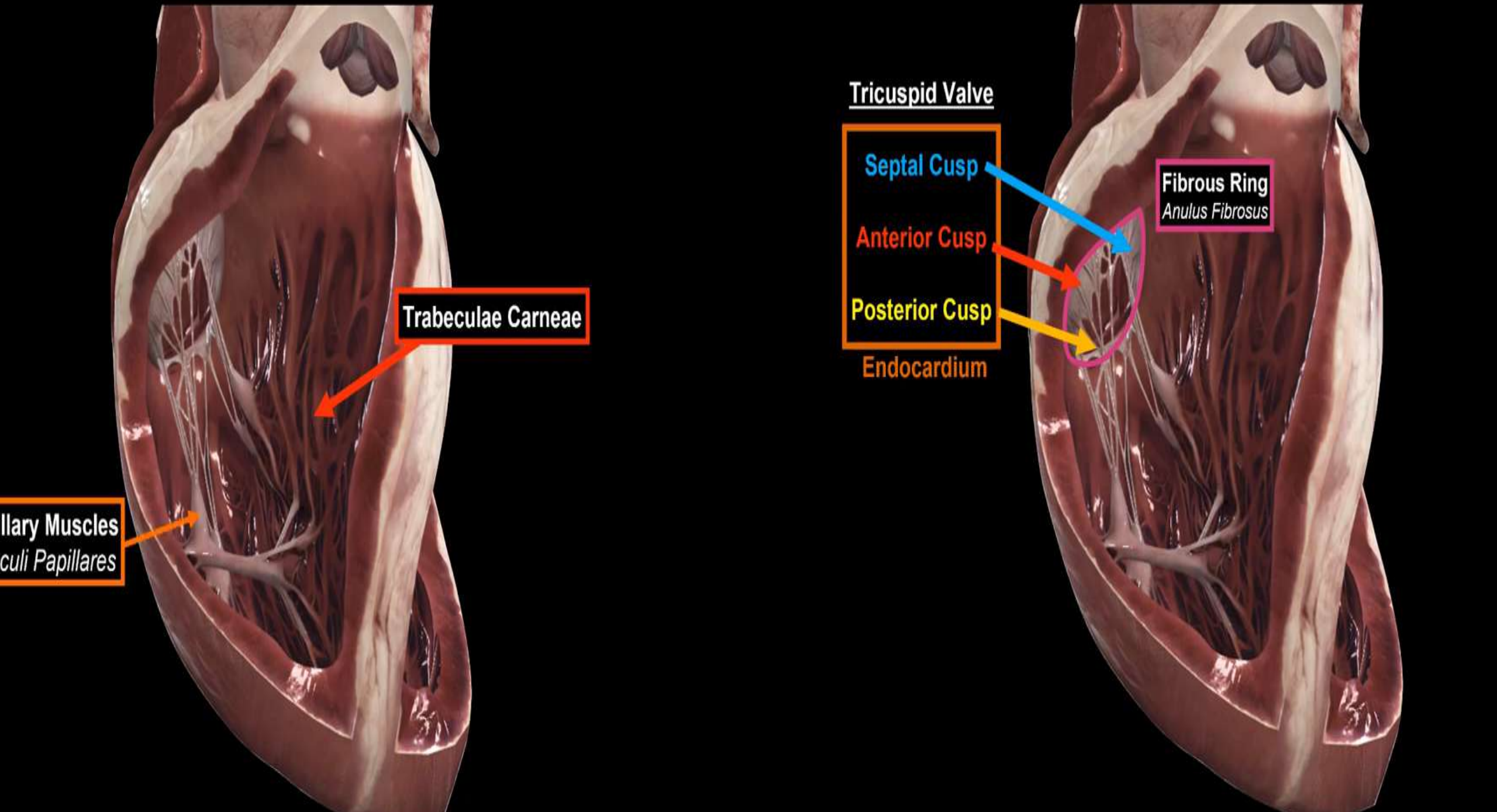
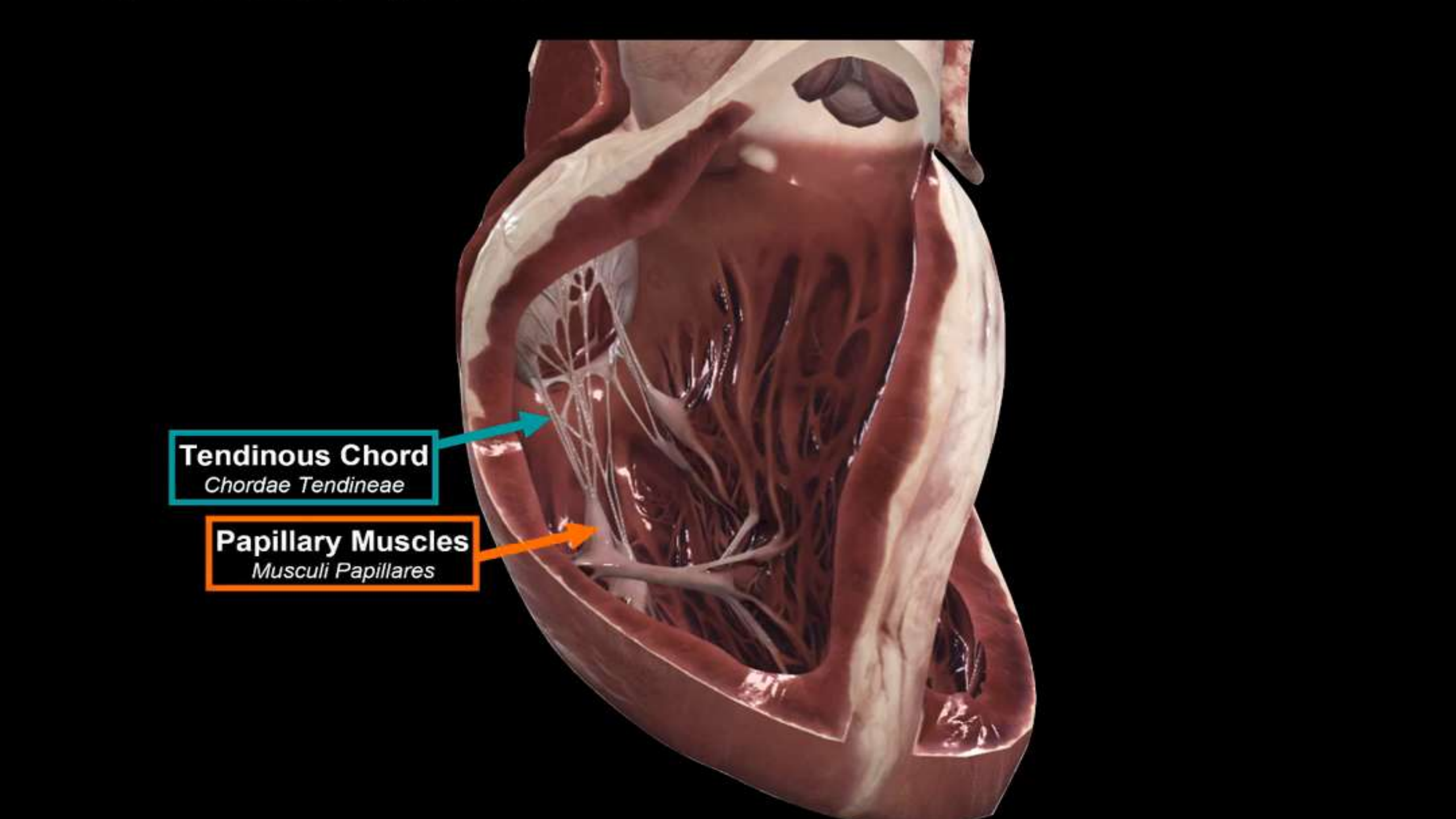


Fig. 3.73 Internal View of the Right Ventricle.

Internal Structures of Right Ventricle



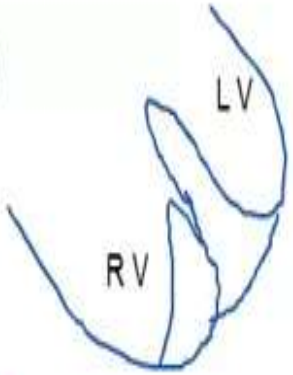

Tricuspid Valve
Septal Cusp
Anterior Cusp
Posterior Cusp
Endocardium
Fibrous Ring
Anulus Fibrosus
Trabeculae Carneae
llary Muscles
culi Papillares


Tendinous Chord
Chordae Tendineae
Papillary Muscles
Musculi Papillares

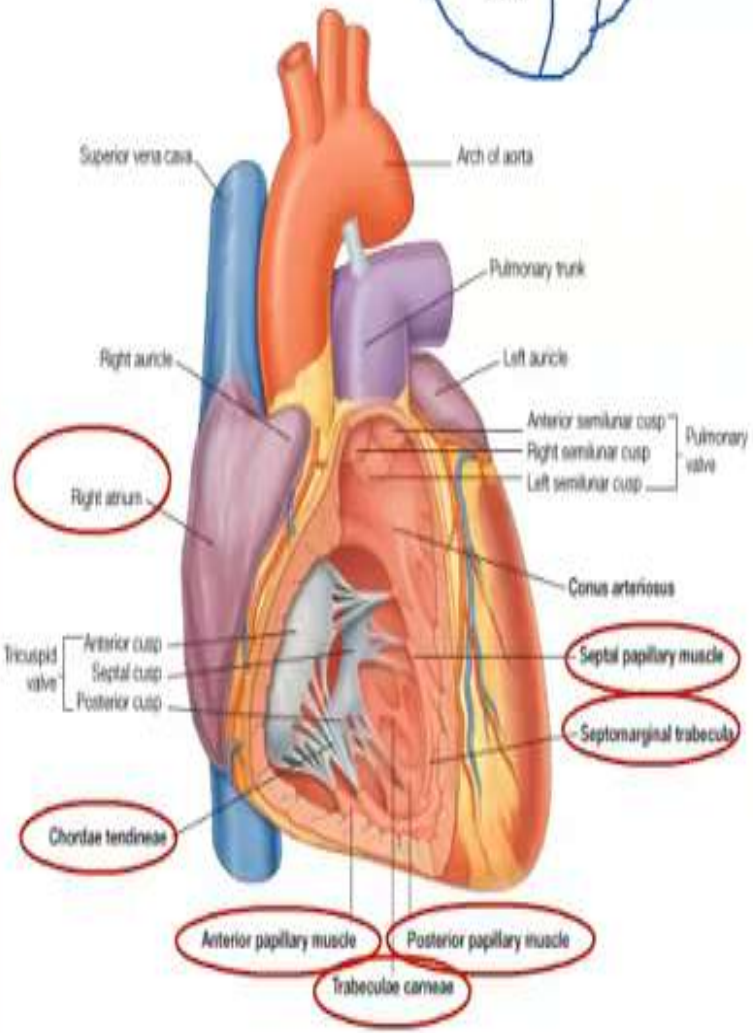
Right ventricle: it forms.....of the heart?

Interior of right ventricle

Rough part

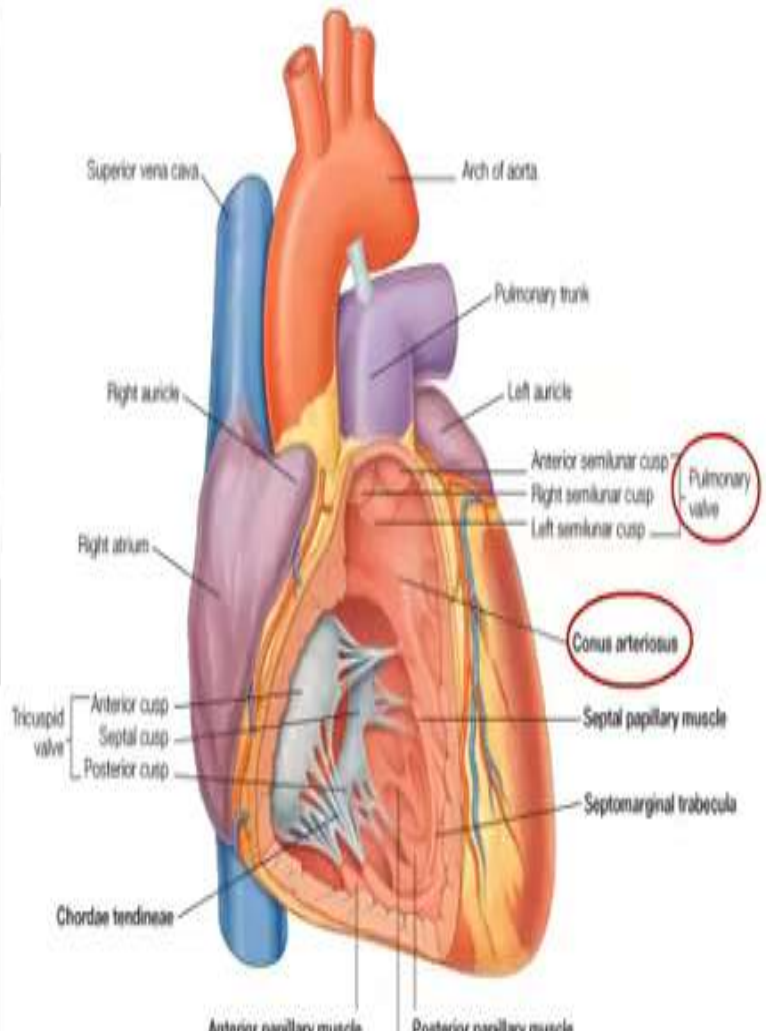


- It is the inflowing part.
- It shows muscular ridges giving the sponge shape (trabeculae carni) which may be:
- Fixed ridges.
- Pyramidal with attached base and free apex (papillary muscles): they are three, one for each cusp of tricuspid valve.
- Attached from both ends and free in between: One of them is the septomarginal band



Interior of right ventricle

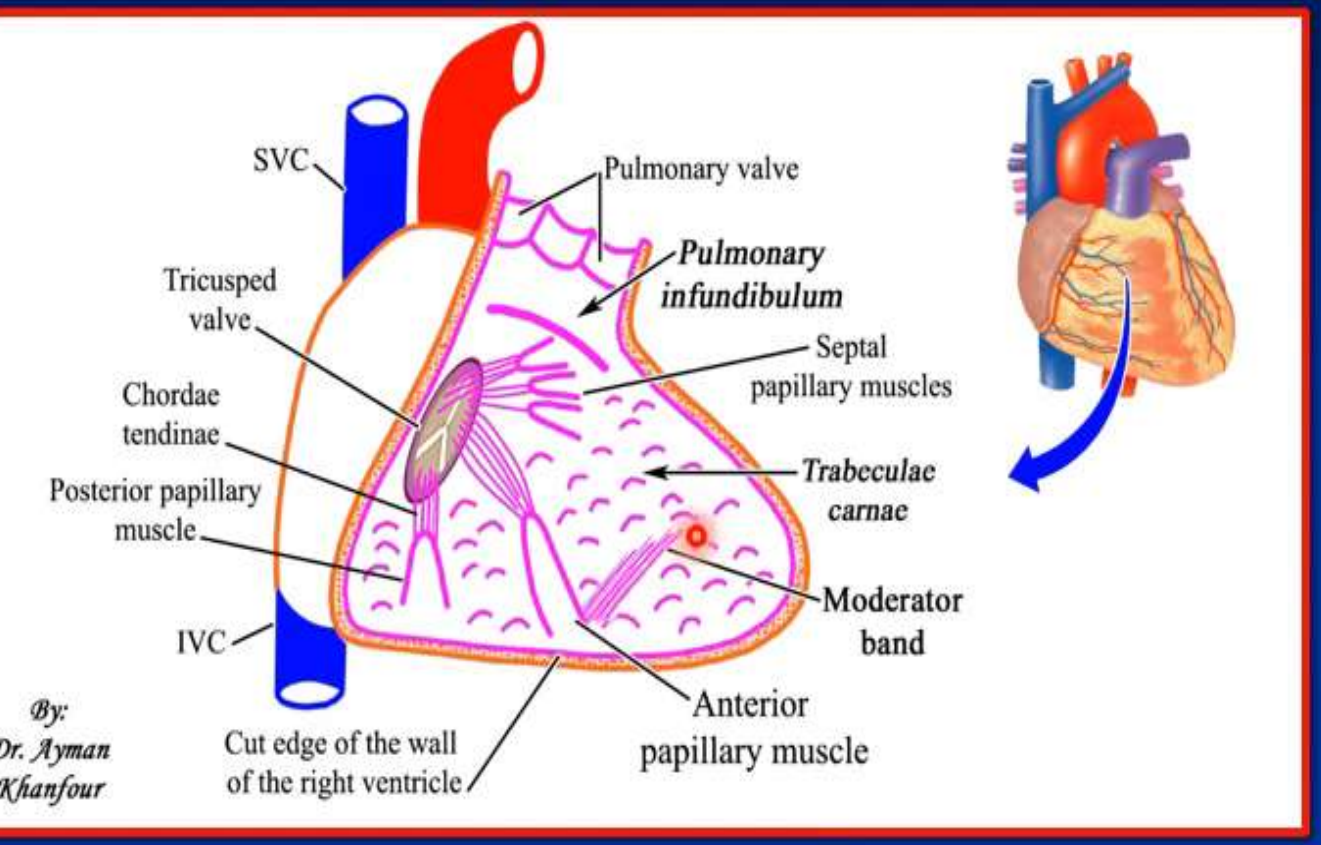
Rough part



S

- It is the o
leading to
orifice whi
the pulmo
- It is smoo
shape, so i
(infundibu
arteriosus)

Right Ventricle



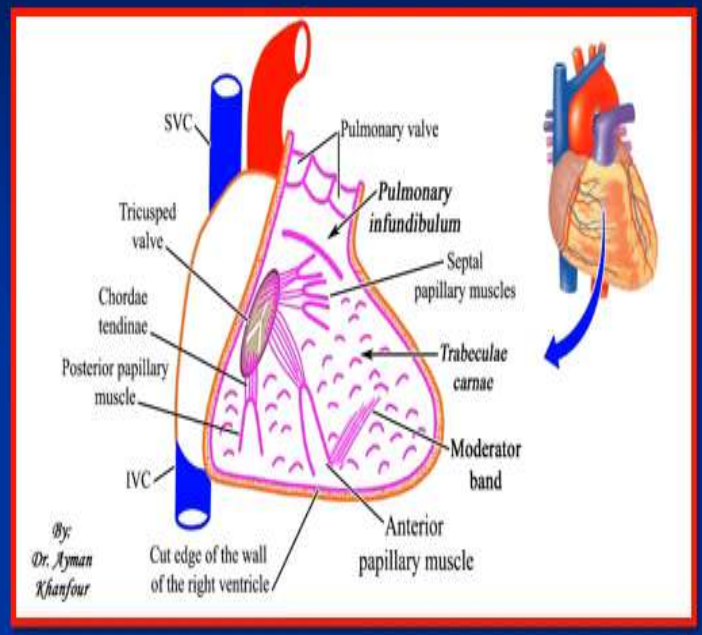
Structures

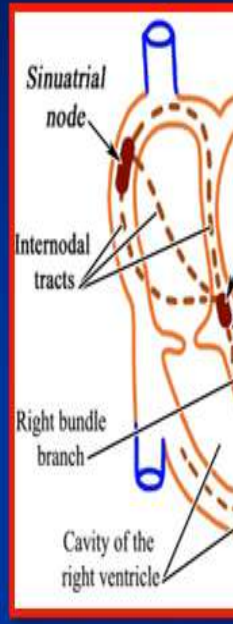
- Trabeculae carnae: It forms rough part of Lt. vent.
- Papillary muscles: Finger like muscular projections.
- Chorda tendineae: Thread like projections that connect the apex of the papillary muscles to the

s

t (Rough part)
e carnae)
act (Smooth part)
y infundibulum)

- Moderator band (Septomarginal trabecula





- It is an enlarged trabecula of muscular tissue which c
septal wall to the *base of the anterior papillary musc*

Function:

a. It prevents over distention of the right ventricle.
b. It also carries the *right branch of atrio-ventricula*
(of the conducting system of the heart) to anterior papil

Left Ventricle

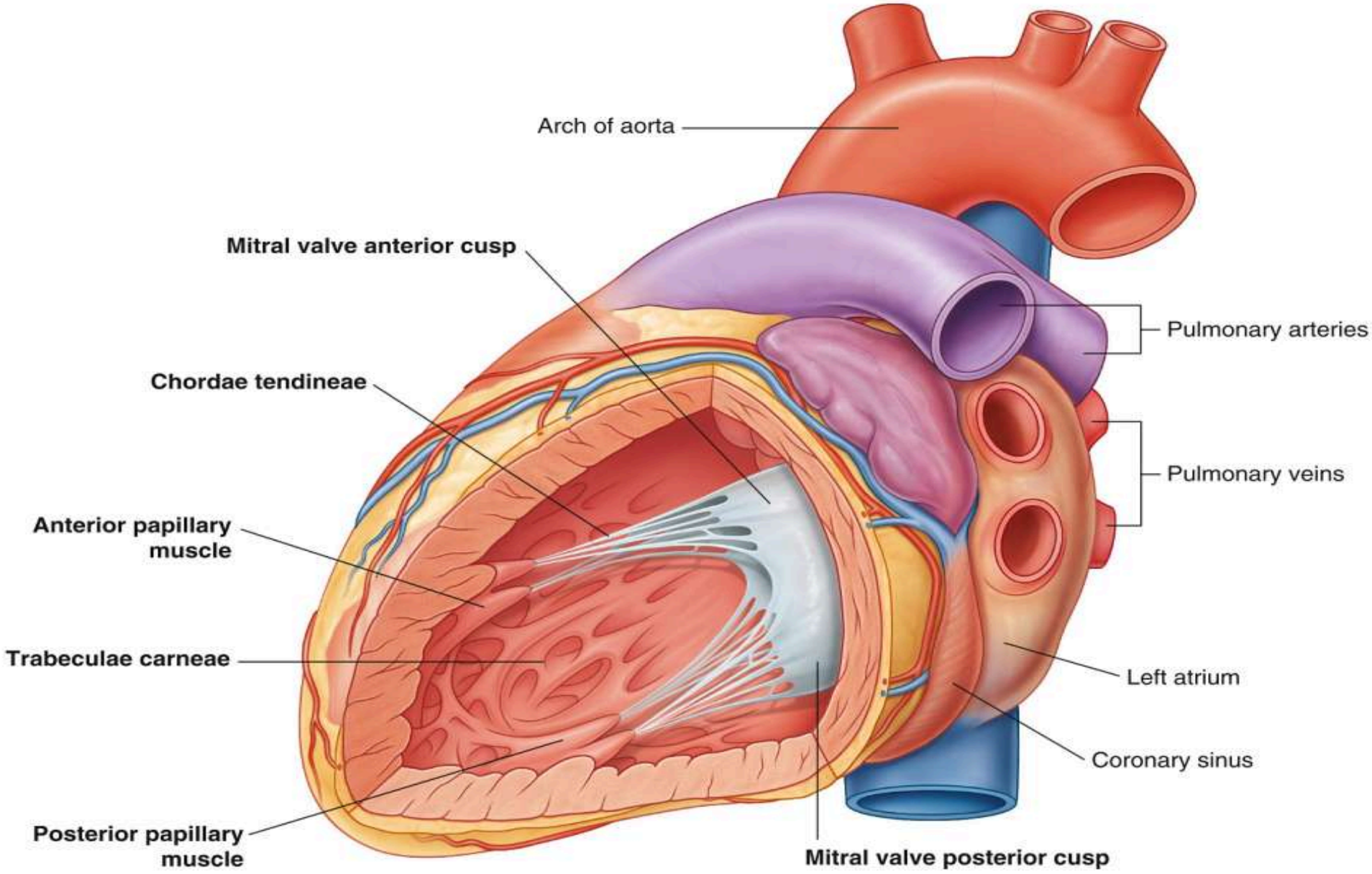
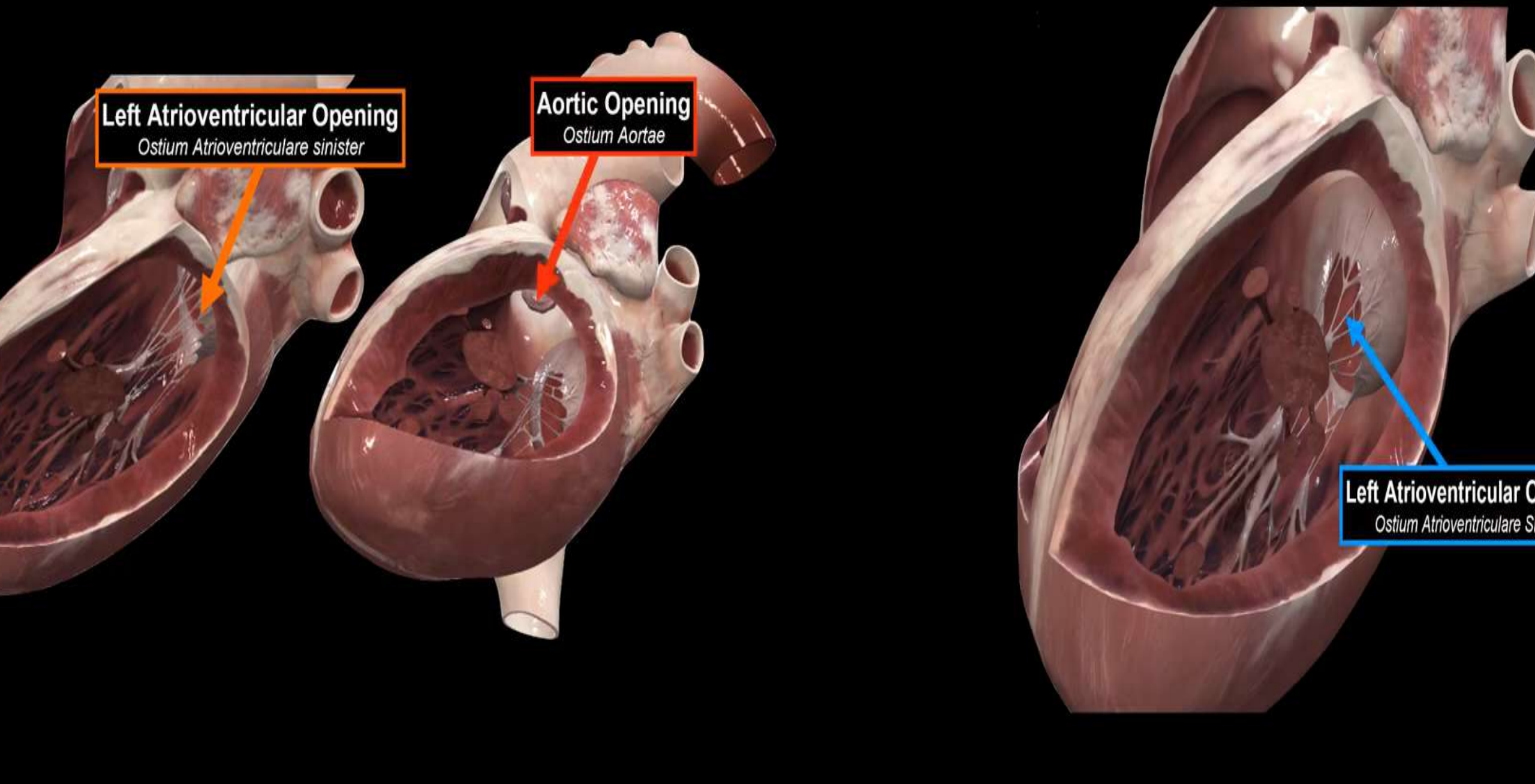


Fig. 3.76 Internal View of the Left Ventricle.


Left Atrioventricular Opening
Ostium Atrioventriculare sinister
Aortic Opening
Ostium Aortae
Left Atrioventricular O
Ostium Atrioventriculare S

entricle: it forms.....of the heart?

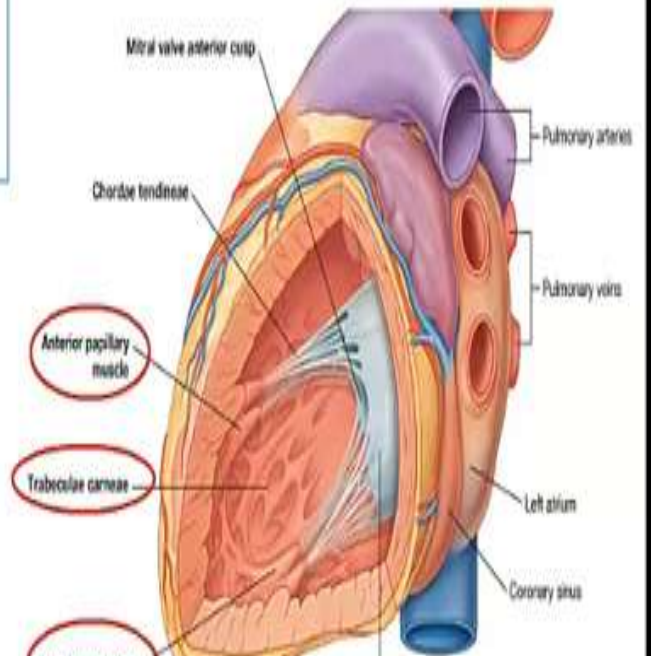
Interior of left ventricle

Rough part

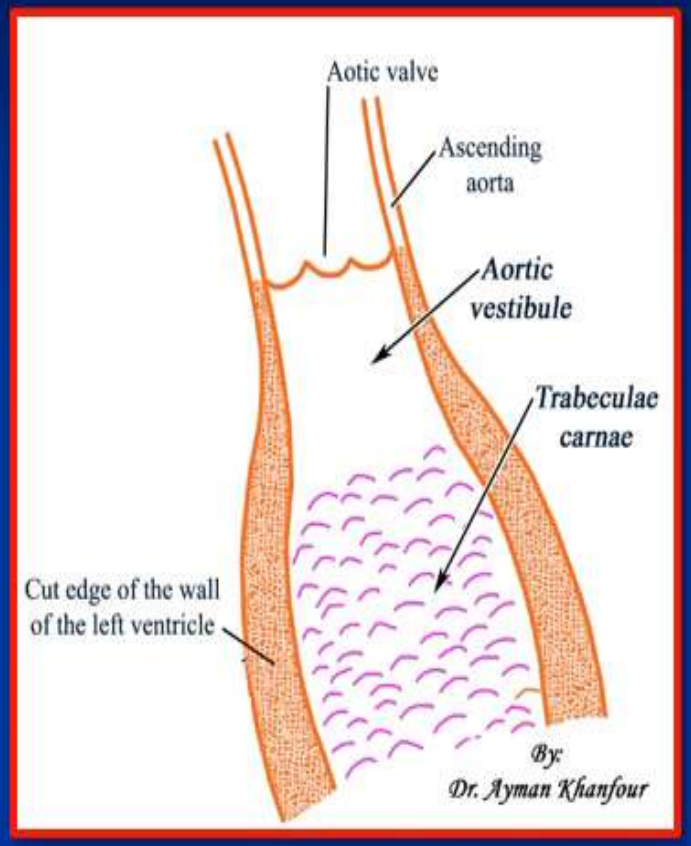
s the inflowing part.
beculae carni are more branching
l finer than that of right
tricle.
y 2 papillary muscles, one for
h cusp of the mitral valve.

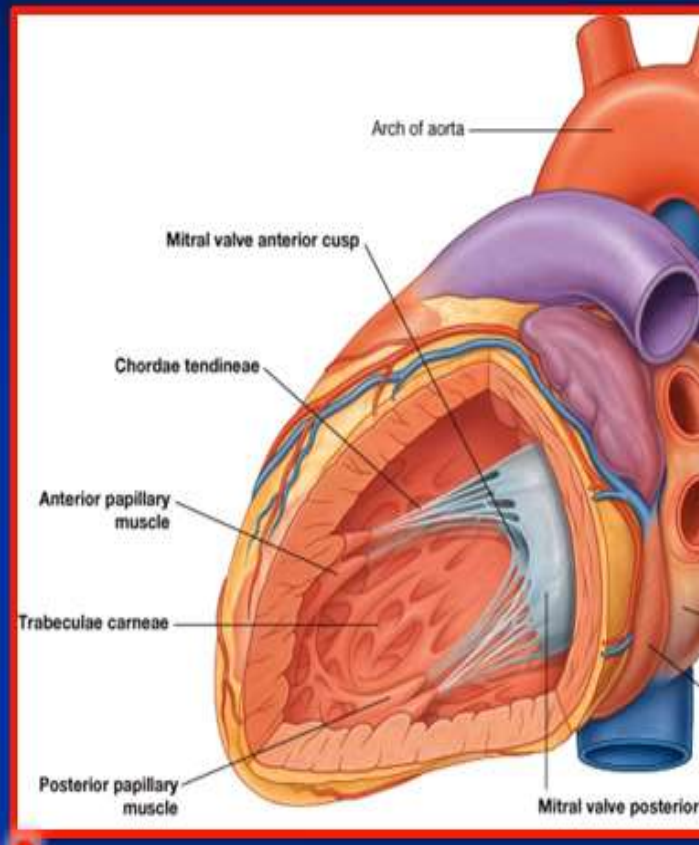
Smooth part

- It is the outflowing part.
- It is called aortic vestibule leading to the aortic orifice.



Left Ventricle





Parts

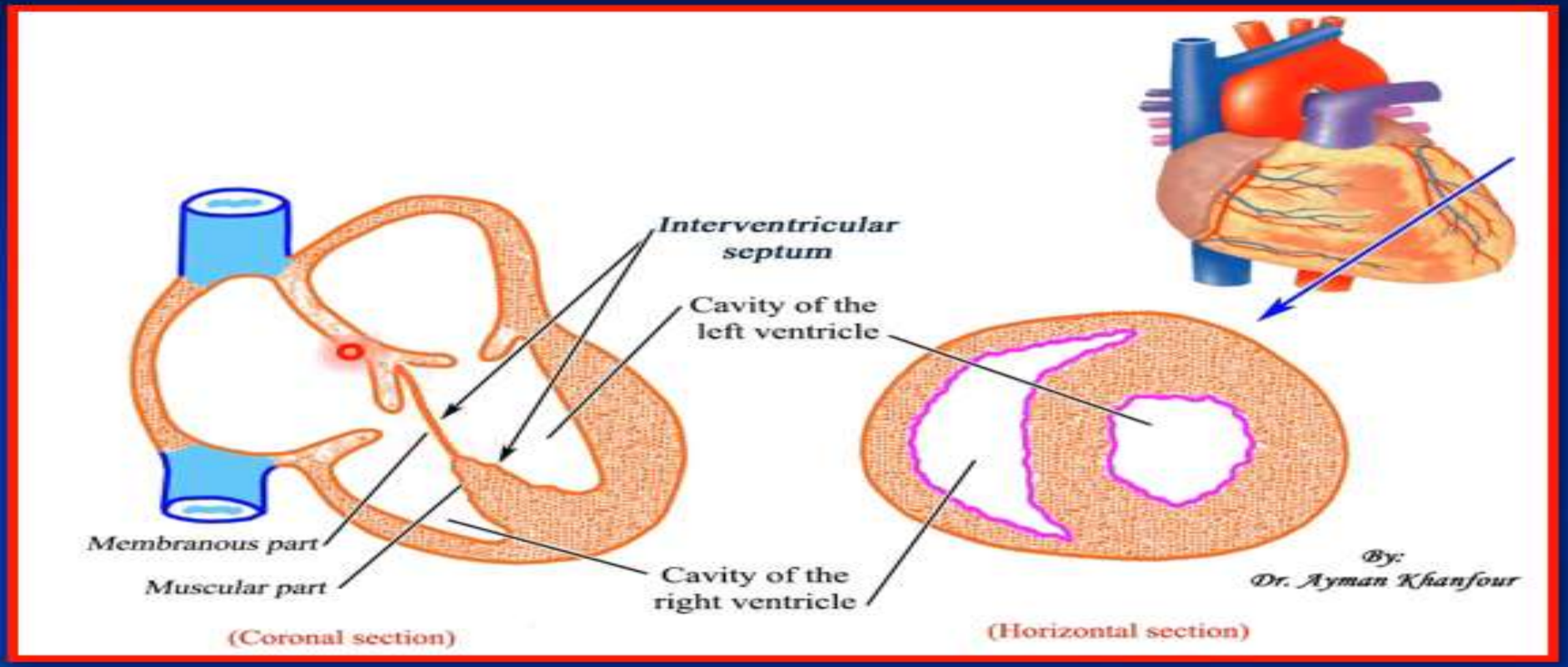
1. Inflow tract (Rough part) (Trabeculae carnae)
2. Outflow tract (Smooth part) (Aortic vestibule)

Structures

- Trabeculae carnae: It forms rough pa
- Papillary muscles: Finger like muscul
- Chorda tendineae: Thread like proje connect the apex of the papillary mu

Interventricular Septum

Interventricular Septum

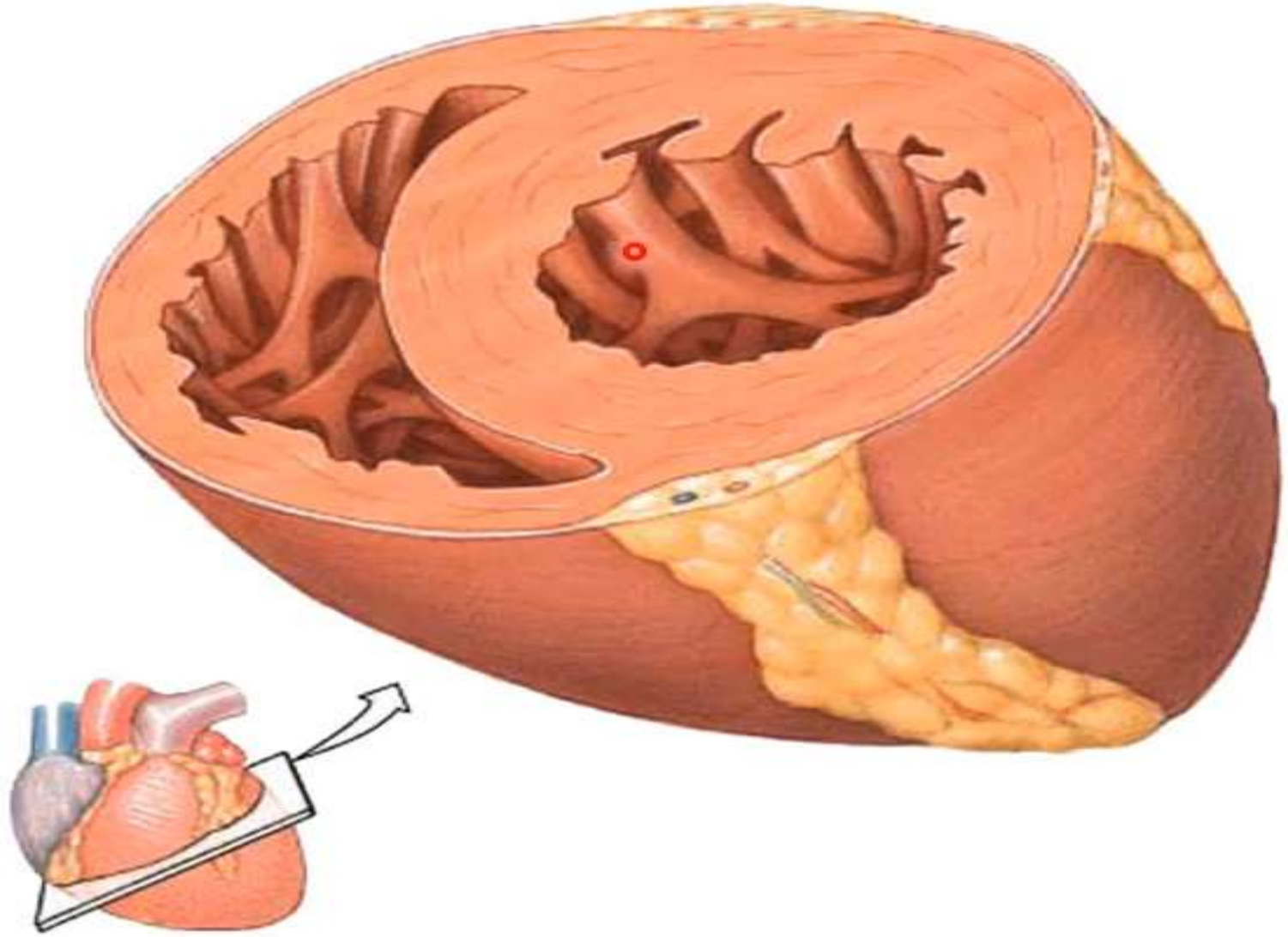


Muscular part

It forms the lower and larger part of the septum.

Membranous part

It forms the upper and smaller part of the septum.



(Interventricular septum)

THANKS